

École Doctorale de Sciences Mathématiques de Paris-Centre (ED 386)
Université Pierre et Marie Curie - Sorbonnes Universités

HABILITATION À DIRIGER DES RECHERCHES

Spécialité “Mathématiques Appliquées”

GEOMETRIC CONTROL OF QUANTUM AND COMPLEX SYSTEMS

présentée et soutenue publiquement par

Marco Caponigro

le 8 Novembre 2017

Jury

<i>Rapporteurs</i>	Karine BEAUCHARD	ENS Rennes
	Nicola BELLOMO	King Abdulaziz University
	José Antonio CARRILLO	Imperial College London
<i>Examineurs</i>	Jean-Baptiste CAILLAU	Université Côte d’Azur
	Jean-Michel CORON	UPMC - Sorbonnes Universités
	Thierry HORSIN	Conservatoire National des Arts et Métiers
	Hans-Rudolf JAUSLIN	Université de Bourgogne Franche-Comté
	Pierre ROUCHON	Mines - ParisTech
	Enrique ZUAZUA	Universidad Autónoma de Madrid

Contents

Preface	i
1 Quantum control	1
1.1 Introduction	1
1.1.1 State of the art	2
1.1.2 Contributions	3
1.1.3 Contents	4
1.2 Bilinear Schrödinger equation	4
1.2.1 Abstract framework	4
1.2.2 Existence	5
1.2.3 Examples	7
1.2.4 Obstructions to exact controllability	8
1.3 Approximate controllability	9
1.3.1 The Lie–Galerkin Condition	9
1.3.2 Results	10
1.3.3 Examples and extensions	11
1.4 Weakly-coupled systems	13
1.4.1 The Good Galerkin Approximation	15
1.4.2 A Bang-Bang theorem for quantum control	16
1.5 Perspectives	16
2 Complex systems	19
2.1 Introduction	19
2.1.1 First-order models	20
2.1.2 Second-order models	21
2.1.3 External control	23
2.1.4 Contributions	25
2.1.5 Contents	26
2.2 General results in sparse control	26
2.2.1 Sparse stabilization of dissipative systems	26
2.2.2 Sparse control of Mean-Field equations	29
2.3 Examples and extensions	33
2.3.1 Global stabilization and optimal control of the Cucker–Smale model	33
2.3.2 Optimal control via leader of social dynamics	35
2.4 Avoiding chattering in optimal control	40
2.5 Modeling	43
2.5.1 Opinion formation on a manifold	43
2.5.2 Psychological bias in pedestrian motion	44
2.6 Perspectives	44

Bibliography	47
References by Marco Caponigro	47
Other References	48

Preface

This *mémoire* contains a summary of my scientific activity since my PhD in 2009. My research focuses on geometric methods in control theory and on their applications, in particular, to the control of quantum and of complex systems.

I started working on the field of quantum control during my first postdoc at Inria-Nancy. I used a twofold approach for the control problem for the discrete-spectrum bilinear Schrödinger equation. On one hand, I studied geometric techniques for the approximate controllability. We introduced a generic sufficient condition, the *Lie–Galerkin Condition*, based on Lie-algebraic finite-dimensional conditions combined with a Galerkin method. One of the main advantages of this control technique is that the results apply to very general systems and in particular to those with strong spectral degeneracies and geometrically non-trivial state space. Moreover, the control laws are, in some case, explicit. On the other hand, I focused on energy estimates for the solutions leading to the notion of *weakly-coupled systems*, which are systems whose evolution is close to the evolution of a suitable finite-dimensional Galerkin approximation. One of the great advantages coming from the regularity of weakly-coupled systems is that it allows extensions of finite-dimensional geometric control techniques to the infinite-dimensional framework. The results on this topic are the outcome of an ongoing collaboration with a team composed by Ugo Boschain, Nabile Boussaïd, Thomas Chambrion, and Mario Sigalotti. Chapter 1 contains an introduction to the problem and a summary of our contributions to the field.

Since 2012, during my stay as a postdoc in an interdisciplinary center at Rutgers-University, I started working on the fascinating topic of complex systems. In this *mémoire* I address the problem of controlling complex systems, more precisely the problem of steering a system of several interacting agents to the equilibrium with a control subject to constraints on its components.

We introduced the notion of *sparse* control: a feedback control steering the system to the equilibrium with at most one non-zero component. The idea is that it is possible to act on a system and enforce the emergence of collective behaviors by acting only on one agent at each instant of time. Sparsity of the control is promoted with a ℓ^1 constraint, a technique we borrowed from image analysis. At the same time, we studied optimal control problems for multiagent system with a *leader*, within the framework of the PhD thesis of S. Wongkaew I co-supervised with Alfio Borzì (thesis defended on October 5, 2015 at Würzburg University). The two approaches, sparse stabilization and control via leader, share common features and, recently, we extended sparse control results to general control-affine systems with a global Lyapunov function, i.e. to general dissipative systems. Thanks to a mean-field limit procedure we adapted these results to transport equations with non-local terms providing a control method for kinetic multiagent models.

We also studied the problem of the emergence of chattering phenomena in general non-linear optimal control problems. Chattering, indeed, naturally arises in the field of multiagent systems, especially when dealing with sparse controls of large groups.

Problems coming from the field of complex systems are subjects of an ongoing collaboration with, among others, Alfio Borzì, Massimo Fornasier, Anna Chiara Lai, Benedetto Piccoli, Francesco Rossi, Emmanuel Trélat. The results are presented in Chapter 2.

The last section of Chapter 2, Section 2.5, shows two results on the modeling of complex systems. Despite a wide literature, mathematical models with their simple dynamics are far from catching the complex nature of multiagent systems. The analysis of non-linear models on Riemannian manifolds or the introduction of behavioral features in mathematical models represent relatively new tools for filling this gap. I decided to present these results very briefly since, although based on geometric methods, they do not concern control.

The fundamental aspect of my research is to study controllability properties that are based on the underlying geometric structure of the problem. This allows to obtain very general and intrinsic results.

This *mémoire* consists of two chapters, one for each field of application of geometric methods: the first, Chapter 1, on quantum control and the second, Chapter 2, on complex systems. The structure of the two chapters is similar. An introduction with the statement of the problem, the motivation for our research, a literature review, and a list of our contributions. The introductions end with a contents section Section 1.1.3 and Section 2.1.5, presenting the detailed structure of the chapter. Then, I present the original contributions following a top-down approach: Starting from the more general one and introducing the other results as examples and corollaries. The presentation hence follows a reverse chronological order. I conclude every chapter with a section on open problems.

The bibliography splits into two sections. References including myself as author, numbered from [1] to [23], are highlighted in the text with the bold brackets **[xx]**. All other references, numbered from [24] on, are contained in a second section and listed in alphabetical order.

Quantum control

1.1 Introduction

The goal of quantum control theory is to design efficient population transfers between quantum states. This task is crucial in atomic and molecular physics, with applications ranging from photochemistry to quantum information, and has attracted increasing attention among quantum physicists, chemists, computer scientists, and control theorists. In order to design controls influencing the system, the classical tools from control theory need to be adapted to the specific features of the quantum world. Some of these features are directly taken into account in the dynamical model, as for instance the monotonicity of the norm, its conservation for closed systems (coming from the probabilistic nature of wavefunctions), or the representation of the state as a density matrix. Other features should be taken into account in the control design: the specific consequences of performing a measurement of a quantum system, for instance, make quantum feedback a completely new domain of research with respect to classical control theory [129]. Similarly, the boundary between the quantum realm and the macroscopic classical world, and the related notion of decoherence, make it crucial to apply efficient control strategies. Applying too large controls for too long can lead, indeed, to a falsification of the model describing the system. We refer to the monograph [71], see also the review [126], for a primer on quantum control theory.

A great variety of models in quantum control have been proposed, ranging from low-dimensional conservative ODEs (e.g., descriptions of coherent single spin dynamics) to dissipative systems of PDEs (modeling, for instance, open systems with infinitely many energy levels). The choice of the model depends on the specific physical system under consideration. In this *mémoire* we adopt a general point of view and we consider the controllability problem for the Schrödinger equation

$$i\frac{d\psi}{dt}(t) = (H_0 + u_1(t)H_1 + \dots + u_p(t)H_p)\psi(t), \quad (1.1)$$

where ψ belongs to the Hilbert sphere of a complex Hilbert space $\mathcal{H}$ and $H_0, \dots, H_p$ are self-adjoint operators on $\mathcal{H}$. The control functions $u_1(\cdot), \dots, u_p(\cdot)$, representing the intensity of the external fields, are real-valued functions of time. The reference model is the one in which $\mathcal{H} = L^2(D, \mathbb{C})$ for some domain $D \subset \mathbb{R}^n$, $H_0 = -\Delta + V(x)$, $H_j = W_j(x)$, $j = 1, \dots, p$, and $V, W_1, \dots, W_p$ are real-valued functions (identified with the corresponding multiplicative operators) characterizing respectively the autonomous dynamics and the coupling of the system with the control u . However, equation (1.1) can be used to describe more general controlled

dynamics. For instance, a quantum particle on a Riemannian manifold subject to an external field (in this case Δ is the Laplace–Beltrami operator) or a two-level ion trapped in a harmonic potential (the so-called Eberly–Law Model [102, 75, 49]). In the last case, as in many others relevant physical situations, the operator H_0 cannot be written as the sum of a Laplacian plus a potential.

Equation (1.1) is usually called *bilinear Schrödinger equation* in the control community, the term *bilinear* referring to the linear dependence with respect to ψ and the affine dependence with respect to u . (The term linear is reserved for systems of the form $\dot{x} = Ax + Bu(t)$.) The operator H_0 is usually called the *drift Hamiltonian* while $H_1, \dots, H_p$ are the control operators. We speak of finite-dimensional or infinite-dimensional systems depending on the dimension of $\mathcal{H}$.

The controllability problem consists in establishing whether, for every pair of states ψ_0 and ψ_1 , there exist a control $u(\cdot)$ and a time T such that the solution of (1.1) with initial condition $\psi(0) = \psi_0$ satisfies $\psi(T) = \psi_1$.

1.1.1 State of the art

The controllability of finite-dimensional systems is a well-established topic (see for instance [71] and references therein), thanks to general controllability methods for left-invariant control systems on compact Lie groups [55, 95, 93, 80, 74].

For infinite-dimensional systems the situation is complicated by the subtleties of the evolution in Banach spaces and the consequent fragmental nature of controllability theory for PDEs. For this reason, most of the current literature concentrates on closed (i.e., conservative) systems. A first series of results that have been obtained concerns obstructions to exact controllability [31, 141]. They basically state that if the control Hamiltonians are bounded as operators on the infinite-dimensional space where the system is set, then the set of states attainable from any given initial condition has empty interior. Moreover, it has been shown that one of the most natural infinite-dimensional models with discrete spectrum, the quantum oscillator, is not even approximately controllable [106]. The first positive controllability result is given in [35, 36], and it has been then refined and extended in [38]. The result is highly non-trivial and it is based on Coron’s return method (see [64]). Exact controllability is proven to hold among regular enough wavefunctions. In particular, exact controllability among eigenfunctions of the uncontrolled Schrödinger operator can be achieved. It should be mentioned that these exact controllability results are for now restricted to Schrödinger equations on 1D domains. Other important approximate controllability results have then been proved using Lyapunov methods [90, 105, 112, 113], both in the case of discrete and mixed spectrum. Methods based on non-holonomicity and computation of iterated Lie brackets are difficult to apply in the infinite-dimensional setting but can provide some results for systems with bounded Hamiltonians [48] or having a sequence of finite-dimensional subspaces which are invariant for some part of the dynamics [49]. This kind of results apply to models of trapped ions (see also [75]). Optimal control approaches [52, 90] and analytical tools for the control of linear systems [144, 33, 34] have also been considered.

1.1.2 Contributions

In the last years we introduced and developed an original method for the controllability of (1.1), based on a fine study of the control properties of the Galerkin approximations of the original infinite-dimensional system. The first result in this direction was obtained in [61], in the form of sufficient conditions for approximate controllability. The result has been extended to systems with spectral resonances [2] from which we developed a constructive algorithm for the design of efficient controls. The constructive algorithms gave rise to the software *Q-track*, developed in Scilab and registered in 2011 at the French Program Protection Agency (APP). In [3] we improved the result by exploiting the non-commutativity in multi-input systems and we introduced a generic sufficient condition for approximate controllability, the *Lie–Galerkin Condition*. The name *Lie–Galerkin* comes from the idea that it is possible to successfully combine a Lie-algebraic finite-dimensional condition with the Galerkin method to find sufficient conditions for approximate (simultaneous) controllability. Indeed the underlying idea of this Lie–Galerkin technique is to choose a suitable sequence of Galerkin approximations, then to use finite-dimensional geometric control tools in order to prove *strong controllability* (in some suitable sense) for each Galerkin approximation, and finally to show how these controllability properties provide approximate controllability for the original infinite-dimensional system.

Recently, in [21] we refined former approximate controllability results by proving that the Lie–Galerkin Condition provides approximate controllability with very regular (i.e. smooth) controls. Moreover, if the end-point mapping, associating with every control the corresponding solution at time T , is continuous (for a suitable topology), then one can control exactly in projections on the span of the first eigenstates. Namely, it has been shown that, for every dimension n of the Galerkin approximation and every initial and target condition, there exists a control steering in finite time the infinite-dimensional control system from the initial condition to a final state having the same first n coordinates as the target. In quantum systems, this problem is quite natural, since in physical experiments one can measure with accuracy only the low-energy states. From the mathematical viewpoint, this kind of study is particularly interesting since the continuity of the end-point mapping is an obstacle to exact controllability, hence this approximate controllability result is the strongest one can expect. Exact controllability in projection has been introduced and proved using geometric techniques based on controllability result for the Galerkin approximations in [24] for the 2D Euler and Navier–Stokes equations (see [133] for the 3D case).

A great advantage of our methods with respect to the results in [49] is that they do not require any invariance assumption. Moreover the Lie–Galerkin technique applies to a large range of problems (in particular to Schrödinger equations on domains and manifolds of any dimension). In addition, the controls achieving controllability can be chosen to be small in magnitude, uniformly bounded in L^1 -norm, regular, and in some cases explicitly computable. The smallness and the regularity of the controls are important to ensure the pertinence of the model. In this moment our approach looks to be the most suitable to obtain positive approximate controllability results for systems on manifolds with non-trivial topology or with several spectral resonances. An important functional-analysis extension of the Lie–Galerkin technique is the notion of *weakly-coupled system* introduced in [4]. A crucial feature of these

systems is that they can be precisely approached by finite-dimensional Galerkin approximations. This leads to controllability analysis through the so-called *Good Galerkin Approximation* and allows to extend finite-dimensional results to the infinite-dimensional setting.

1.1.3 Contents

In Section 1.2 we introduce the abstract bilinear Schrödinger equation. We define the solution in Section 1.2.1, we discuss the well-posedness and we determine the admissible classes of controls in Section 1.2.2. In Section 1.2.3 we provide examples of quantum systems falling within our framework. In Section 1.2.4 we discuss the obstructions to the exact controllability and present an extension of the well-known result by Ball–Marsden–Slemrod to the L^1 case. This will serve as motivation for the introduction of the notion of approximate controllability.

In Section 1.3 we present our results in approximate controllability. More precisely we define the Lie–Galerkin Condition in Section 1.3.1 and we state the main results in controllability in Section 1.3.2. Applications of these results, as well as their corollaries and extensions, are presented in Section 1.3.3.

In Section 1.4 we introduce the notion of weakly-coupled systems. These are systems which are sufficiently regular for the propagator to be close to the propagator of the finite-dimensional Galerkin approximations. The existence of this finite-dimensional approximation is presented in Section 1.4.1 and in Section 1.4.2 we present an extension of a finite-dimensional bang-bang result to the infinite-dimensional setting.

We conclude, in Section 1.5, with a description of possible evolutions of these works and open problems in the field.

1.2 Bilinear Schrödinger equation

1.2.1 Abstract framework

In stead of the operators $H_0, \dots, H_p$ in (1.1) we consider, for simplicity, the bilinear control systems obtained by setting $A = -iH_0$ and $B_j = -iH_j$, $j = 1, \dots, p$. Let $\mathcal{H}$ be an infinite-dimensional Hilbert space with scalar product $\langle \cdot, \cdot \rangle$ and $A, B_1, \dots, B_p$ be (possibly unbounded) skew-adjoint operators on $\mathcal{H}$, with domains $D(A), D(B_1), \dots, D(B_p)$ respectively. We consider controls $u = (u_1, \dots, u_p)$ belonging to a set U of control parameters of the form $U = U_1 \times \dots \times U_p$, where, for some $\delta > 0$, either $U_j = [0, \delta]$ or $U_j = [-\delta, \delta]$ depending on the physics of the problem. We consider the controlled equation

$$\frac{d\psi}{dt}(t) = (A + u_1(t)B_1 + \dots + u_p(t)B_p)\psi(t), \quad u(t) \in U, \quad (1.2)$$

with initial condition $\psi(0) = \psi_0 \in \mathcal{H}$ with $\|\psi_0\| = 1$.

We say that A satisfies (A1) if the following assumption holds true.

(A1) A has discrete spectrum with infinitely many distinct eigenvalues (possibly degenerate).

Note that (A1) is true whenever A has compact resolvent. Denote by Φ a Hilbert basis $(\phi_k)_{k \in \mathbb{N}}$ of $\mathcal{H}$ made of eigenvectors of A associated with the family of eigenvalues $(i\lambda_k)_{k \in \mathbb{N}}$ and let $\mathcal{L}$ be the set of finite linear combinations of eigenstates, that is,

$$\mathcal{L} = \bigcup_{k \in \mathbb{N}} \text{span}\{\phi_1, \dots, \phi_k\}.$$

We consider the following assumptions:

(A2) $\phi_k \in D(B_j)$ for every $k \in \mathbb{N}, j = 1, \dots, p$;

(A3) $A + u_1 B_1 + \dots + u_p B_p : \mathcal{L} \rightarrow \mathcal{H}$ is essentially skew-adjoint for every $u \in U$.

Assumption 1.1. The skew adjoint operator A satisfies (A1) and the tuple $(A, B_1, \dots, B_p, \Phi, U)$ satisfies (A2) – (A3)

Assumption 1.1 is the minimal setting for the definition of solution of (1.2) which is given as follows.

Definition 1.1. We say that $u \in L^\infty([0, T], \mathbb{R}^p)$ is *admissible* for (1.2) if $u(t) \in U$ for almost every $t \in [0, T]$ and, for every $\psi_0 \in \mathcal{H}$, there exists $\psi : [0, T] \rightarrow \mathcal{H}$ such that $\psi(0) = \psi_0$ the function $t \mapsto \langle \phi_k, \psi(t) \rangle$ is absolutely continuous for every $k \in \mathbb{N}$ and satisfies

$$\frac{d}{dt} \langle \phi_k, \psi(t) \rangle = -\langle (A + u_1(t)B_1 + \dots + u_p(t)B_p) \phi_k, \psi(t) \rangle, \quad (1.3)$$

for almost every $t \in [0, T]$. The function $t \mapsto \psi(t)$ is called *solution* of (1.2) with initial condition $\psi_0 \in \mathcal{H}$ associated with the control u .

Assumption (A3) implies that the norm of the solutions given by Definition 1.1 is constant along the evolution. In particular it guarantees the uniqueness of solutions. We can therefore define the unitary propagator of (1.2).

Definition 1.2. Let $u : [0, T] \rightarrow U$ be admissible for (1.2). The mapping $[0, T] \ni t \mapsto \Upsilon_t^u$ where $\Upsilon_t^u \psi_0$ is the evaluation at time t of the solution of (1.2) with initial condition $\psi_0 \in \mathcal{H}$ associated with u , is called *propagator* of (1.2).

1.2.2 Existence

In the case in which the control operators $B_1, \dots, B_p$ are bounded the solution of (1.2) is well-defined for a very large class of controls.

Proposition 1.1 (Proposition 2.1 in [31]). *If $B_1, \dots, B_p$ are bounded then every Banach space $\mathcal{U}$ continuously and densely included in $L^1([0, T], U)$, for $T > 0$ given, is a class of admissible controls.*

Assumption 1.1 is sufficient to define the solution of (1.2) associated with piecewise constant controls even in the unbounded case.

Proposition 1.2. *Let $(A, B_1, \dots, B_p, \Phi, U)$ satisfy Assumption 1.1 and let $u(\cdot) = (u_1(\cdot), \dots, u_p(\cdot))$ be a p -tuple of piecewise constant controls on an interval $[0, T]$, for $T > 0$ given, with values in U . Then u is admissible for (1.2) and the propagator is given by*

$$\Upsilon_t^u = e^{(t - \sum_{l=1}^{j-1} t_l)(A + u_1^{(j)} B_1 + \dots + u_p^{(j)} B_p)} \circ \dots \circ e^{t_1(A + u_1^{(1)} B_1 + \dots + u_p^{(1)} B_p)},$$

where $\sum_{l=1}^{j-1} t_l \leq t < \sum_{l=1}^j t_l$ and $u(\tau) = (u_1^{(k)}, \dots, u_p^{(k)}) \in U$ if $\sum_{l=1}^{k-1} t_l \leq \tau < \sum_{l=1}^k t_l$.

If, moreover, $D(A) = D(A + u_1 B_1 + \dots + u_p B_p)$ for every $u \in U$ then, given $\psi_0 \in D(A)$, $\Upsilon_t^u(\psi_0)$ satisfies (1.2) for almost every $t \in [0, T]$.

In order to define the solution associated with smooth controls one needs extra regularity assumptions.

Definition 1.3. B is said *relatively bounded with respect to A* , or *A -bounded*, if $D(A) \subset D(B)$ and there exist $a, b > 0$ such that for every ψ in $D(A)$,

$$\|B\psi\| \leq a\|A\psi\| + b\|\psi\|. \quad (1.4)$$

The A -bound of B is the smallest constant a for which (1.4) holds true.

Recall that the set U is bounded, and, in particular $U \subset [-\delta, \delta]^p$ for some $\delta > 0$.

Proposition 1.3. *Let $(A, B_1, \dots, B_p, \Phi, U)$ satisfy Assumption 1.1 and assume that $B_1, \dots, B_p$ are A -bounded with A -bound smaller than $1/\delta$. Then given $T > 0$, every $u \in C^1([0, T], U)$ is admissible for (1.2).*

If, moreover, $D(A) = D(A + u_1 B_1 + \dots + u_p B_p)$ for every $u \in U$ then, given $\psi_0 \in D(A)$, $\Upsilon_t^u(\psi_0)$ satisfies (1.2) for every $t \in [0, T]$.

The first part of the statement is a consequence of the Kato–Rellich theorem ([122, Theorem X.12]) and the second part is [122, Theorem X.70].

It is also possible to extend the notion of solution to BV controls. In this case one has a *mild* solution of (1.2), namely

$$\psi(t) = e^{tA}\psi_0 + \int_0^t e^{(t-s)A}(u_1(s)B_1 + \dots + u_p(s)B_p)\psi(s)ds,$$

for every $t \in [0, T]$.

Proposition 1.4. *Let $(A, B_1, \dots, B_p, \Phi, U)$ satisfy Assumption 1.1 and assume that $B_1, \dots, B_p$ are A -bounded with A -bound smaller than $1/\delta$. Assume moreover that $D(A) = D(A + u_1 B_1 + \dots + u_p B_p)$ for every $u \in U$. Then, given $T > 0$, every $u \in BV([0, T], U)$ is admissible and the associated solution $\psi(\cdot)$ is a mild solution of (1.2).*

This is a consequence of [96] (see also Theorem 5 in [10]). The invariance of the domains, although classical, is quite a strong assumption.

1.2.3 Examples

We present three examples of quantum systems satisfying Assumption 1.1.

Example 1.4 (Quantum harmonic oscillator). The quantum harmonic oscillator is among the most important examples of quantum systems (see, for instance, [63, Complement G_V]). Its controlled version has been extensively studied (see, for instance, [106]). In this example $\mathcal{H} = L^2(\mathbb{R}, \mathbb{C})$ and equation (1.2) reads

$$i \frac{\partial \psi}{\partial t}(x, t) = \frac{1}{2}(-\Delta + x^2)\psi(x, t) + u(t)x\psi(x, t). \quad (1.5)$$

A Hilbert basis of $\mathcal{H}$ made of eigenvectors of $A = -\frac{i}{2}(-\Delta + x^2)$ is given by the sequence of the Hermite functions $(\phi_n)_{n \in \mathbb{N}}$, associated with the sequence $(-i\lambda_n)_{n \in \mathbb{N}}$ of eigenvalues where $\lambda_n = n - 1/2$ for every n in $\mathbb{N}$. In the basis $(\phi_n)_{n \in \mathbb{N}}$, $B_1 = B = -ix$ admits a *tri-diagonal structure* since

$$\langle \phi_j, B\phi_k \rangle = \begin{cases} i\sqrt{k-1} & \text{if } j = k-1, \\ i\sqrt{k} & \text{if } j = k+1, \\ 0 & \text{otherwise.} \end{cases}$$

Example 1.5 (Potential well). One of the first results of exact controllability has been obtained (see [35, 36, 38]) for a system of a quantum particle confined to 1D infinite square potential well and subject to an electric field. The Schrödinger equation reads

$$i \frac{\partial \psi}{\partial t}(x, t) = -\Delta \psi(x, t) - u(t)\mu(x)\psi(x, t), \quad (1.6)$$

with the boundary conditions $\psi(0, t) = \psi(1, t) = 0$ for every $t \in \mathbb{R}$ and $\mu \in H^3((0, 1), \mathbb{R})$. The controls $u(\cdot)$ are piecewise constant with values in $U = [-\delta, \delta]$ for some $\delta > 0$.

In this case $\mathcal{H}$ is the space $L^2((0, 1), \mathbb{C})$. The operators A and $B_1 = B$ are defined by $A\psi = i \frac{\partial^2 \psi}{\partial x^2}$ for every ψ in $D(A) = (H_2 \cap H_0^1)((0, 1), \mathbb{C})$ and $B\psi = i\mu(x)\psi$. A complete set of eigenfunctions of A associated with the eigenvalues $i\lambda_k = -i(k\pi)^2, k \in \mathbb{N}$ is given by

$$\phi_k(x) = \sqrt{2} \sin(k\pi x), \quad k \in \mathbb{N},$$

which form an orthonormal basis of $\mathcal{H}$.

Example 1.6 (Rotating bipolar molecule). We consider the control of the orientation of a rigid linear molecule in $\mathbb{R}^3$ by external fields. The corresponding controlled Schrödinger equation is defined on the unit sphere S^2 . Up to normalization of physical quantities (in particular, in units such that $\hbar = 1$), the dynamics are modeled by the equation

$$i \frac{\partial \psi(\theta, \varphi, t)}{\partial t} = -\Delta \psi(\theta, \varphi, t) + (u_1(t) \sin \theta \cos \varphi + u_2(t) \sin \theta \sin \varphi + u_3(t) \cos \theta) \psi(\theta, \varphi, t), \quad (1.7)$$

where θ, φ are the spherical coordinates, which are related to the Euclidean coordinates through the identities

$$x = \sin \theta \cos \varphi, \quad y = \sin \theta \sin \varphi, \quad z = \cos \theta,$$

while Δ is the Laplace–Beltrami operator on the sphere S^2 (called in this context the *angular momentum operator*), i.e.,

$$\Delta = \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2}.$$

The wavefunction ψ evolves in the unit sphere $\mathcal{S}$ of $\mathcal{H} = L^2(S^2, \mathbb{C})$. A basis of eigenvectors of the Laplace–Beltrami operator Δ is given by the spherical harmonics $Y_\ell^m(\theta, \varphi)$, which satisfy

$$\Delta Y_\ell^m(\theta, \varphi) = -\ell(\ell + 1) Y_\ell^m(\theta, \varphi).$$

The spectrum of $A = i\Delta$ is $\{-i\ell(\ell + 1) \mid \ell \in \mathbb{N}\}$. Each eigenvalue $-i\ell(\ell + 1)$ is of finite multiplicity $2\ell + 1$. The control operators B_1, B_2, B_3 are the multiplication operators by $-i \cos \varphi \sin \theta, -i \sin \varphi \sin \theta, -i \cos \theta$, respectively, which are bounded.

1.2.4 Obstructions to exact controllability

Once well-posedness and existence for (1.2) are assessed we consider the problem of controllability, namely, given a class of admissible controls $\mathcal{U}$, an initial condition ψ_0 , and a final condition ψ_1 in $\mathcal{H}$ with $\|\psi_0\| = 1 = \|\psi_1\|$, does there exist $u \in \mathcal{U}$ and $T > 0$ such that $\Upsilon_T^u(\psi_0) = \psi_1$?

The problem is to describe the *attainable set from ψ_0 with controls in $\mathcal{U}$* , defined as

$$\mathcal{A}(\psi_0, \mathcal{U}) = \bigcup_{T \geq 0} \bigcup_{u \in \mathcal{U}} \{\Upsilon_T^u \psi_0\}.$$

Unfortunately this set has empty interior whenever $\mathcal{H}$ is infinite-dimensional and the control operators are bounded. This is a consequence of the following well-known result by Ball, Marsden, and Slemrod for single-input bilinear systems in Banach spaces.

Theorem 1.5 (Theorem 3.6 in [31]). *Let X a Banach space, $A : D(A) \rightarrow X$ a linear operator generating a C^0 semigroup, and $B : X \rightarrow X$ be bounded. Then, for every ψ_0 in X , the attainable set from ψ_0 with L_{loc}^r controls, $r > 1$, is contained in a countable union of compact subsets of X .*

The idea of the result is that the end-point mapping $u \mapsto \Upsilon_T^u(\psi_0)$ is continuous from $\mathcal{U} = \cup_{r>1} L^r$ endowed with the weak-* topology to X . Hence the attainable set is the image of a countable union of sequentially compact (for the weak-* topology) sets (thanks to Banach–Alaoglu theorem). If X is infinite-dimensional, then the attainable set has empty interior. We recently extended this non-controllability result to L^1 controls in [9]. A major issue here is that the “compactness + continuity” argument does not apply to the L^1 case.

In general, the problem is that the regularity of the control operators is a necessary condition for the well-posedness of the solutions but, on the other hand, whenever the control operators are too regular one may lose exact controllability. For this reason, in the following we focus on weaker controllability properties.

1.3 Approximate controllability

Since exact controllability is not achievable in general we focus our analysis on a weaker controllability property.

Definition 1.7. Let $(A, B_1, \dots, B_p, \Phi, U)$ satisfy Assumption 1.1 and let $\mathcal{U}$ be a class of admissible controls. We say that (1.2) is *approximately controllable* by means of controls in $\mathcal{U}$ if for every ψ_0, ψ_1 in the unit sphere of $\mathcal{H}$ and every $\varepsilon > 0$ there exist $u \in \mathcal{U}$ and $T > 0$ such that $\|\psi_1 - \Upsilon_T^u(\psi_0)\| < \varepsilon$.

1.3.1 The Lie–Galerkin Condition

We present a sufficient condition for the approximate controllability, introduced in [3] and based on a Lie-algebraic condition on the Galerkin approximations of a linear system associated with (1.2). In order to define the so-called Lie–Galerkin Condition we first define for $n \in \mathbb{N}$ the projection Π_n of $\mathcal{H}$ on the span of the first n eigenvectors of A , namely

$$\begin{aligned} \Pi_n : \mathcal{H} &\rightarrow \mathcal{H} \\ \psi &\mapsto \sum_{k=1}^n \langle \phi_k, \psi \rangle \phi_k. \end{aligned}$$

When it does not create ambiguities we identify $\text{Im}(\Pi_n) = \text{span}\{\phi_1, \dots, \phi_n\}$ with $\mathbb{C}^n$.

Given a linear operator Q on $\mathcal{H}$ and a Hilbert basis $\Phi = (\phi_k)_{k \in \mathbb{N}}$ we identify the linear operator $Q^{(n)} = \Pi_n Q \Pi_n$ preserving $\text{span}\{\phi_1, \dots, \phi_n\}$ with its $n \times n$ complex matrix representation with respect to $(\phi_1, \dots, \phi_n)$. The operator $Q^{(n)}$ is called *compression* of Q . The system

$$\dot{x} = Q^{(n)}x, \quad x \in \text{span}\{\phi_1, \dots, \phi_n\},$$

is called *Galerkin approximation* of order n of the infinite-dimensional system

$$\dot{x} = Qx, \quad x \in \mathcal{H}.$$

Let us introduce the set Σ_n of spectral gaps associated with the first n eigenvalues of A as

$$\Sigma_n = \{|\lambda_l - \lambda_k| \mid l, k = 1, \dots, n\}.$$

For every $\sigma \geq 0$, every $m \in \mathbb{N}$, and every $m \times m$ matrix M , let

$$\mathcal{E}_\sigma(M) = (M_{l,k} \delta_{\sigma, |\lambda_l - \lambda_k|})_{l,k=1}^m,$$

where $\delta_{\cdot, \cdot}$ denotes the Kronecker symbol. The $n \times n$ matrix $\mathcal{E}_\sigma(B_j^{(n)})$, $j = 1, \dots, p$, corresponds then to the “selection” in $B_j^{(n)}$ of the spectral gap σ : every element is 0 except the (l, k) -elements such that $|\lambda_l - \lambda_k| = \sigma$.

Define

$$\Xi_n = \{(\sigma, j) \in \Sigma_n \times \{1, \dots, p\} \mid (B_j)_{k,l} \delta_{\sigma, |\lambda_l - \lambda_k|} = 0, \text{ for every } k = 1, \dots, n \text{ and } l > n\}. \quad (1.8)$$

The set Ξ_n can be seen as follows: If $(\sigma, j) \in \Xi_n$ then the matrix $M = \mathcal{E}_\sigma(B_j^{(n)})$ is such that

$$\mathcal{E}_\sigma(B_j^{(N)}) = \left(\begin{array}{c|c} M & 0 \\ \hline 0 & * \end{array} \right) \text{ for every } N > n. \quad (1.9)$$

Definition 1.8 ([3]). For every $n \in \mathbb{N}$ define

$$\begin{aligned} \mathcal{M}_n = & \left\{ A^{(n)} \right\} \cup \left\{ \mathcal{E}_\sigma(B_j^{(n)}) \mid (\sigma, j) \in \Xi_n \text{ and } j \text{ is such that } (0, j) \in \Xi_n \right\} \\ & \cup \left\{ \mathcal{E}_\sigma(B_j^{(n)}) \mid (\sigma, j) \in \Xi_n, \sigma \neq 0, U_j = [-\delta, \delta] \right\}. \end{aligned}$$

We say that the *Lie–Galerkin Condition* holds if for every $n_0 \in \mathbb{N}$ there exists $n > n_0$ such that

$$\text{Lie}\mathcal{M}_n \supseteq \mathfrak{su}(n). \quad (1.10)$$

Here $\text{Lie}\mathcal{M}_n$ is the Lie algebra over $\mathbb{C}$ generated by matrices in $\mathcal{M}_n$. The Lie bracket is the commutator of matrices.

An example of simple verifiable condition in the case of single input systems sufficient for the Lie–Galerkin Condition is given by the existence of a non-resonant chain of connectedness. Let $p = 1$ and call, for simplicity, $B = B_1$.

Definition 1.9 ([2]). Let (A, B, U, Φ) satisfy Assumption 1.1.

A subset S of $\mathbb{N}^2$ couples two levels j, k in $\mathbb{N}$, if there exists a finite sequence $((s_1^1, s_2^1), \dots, (s_1^p, s_2^p))$ in S such that

- (i) $s_1^1 = j$ and $s_2^p = k$;
- (ii) $s_2^j = s_1^{j+1}$ for every $1 \leq j \leq p-1$;
- (iii) $\langle \phi_{s_1^j}, B\phi_{s_2^j} \rangle \neq 0$ for $1 \leq j \leq p$.

S is called a *connectedness chain* for (A, B, U, Φ) if S couples every pair of levels in $\mathbb{N}$.

A connectedness chain is said to be *non-resonant* if for every (s_1, s_2) in S , $|\lambda_{s_1} - \lambda_{s_2}| \neq |\lambda_{t_1} - \lambda_{t_2}|$ for every (t_1, t_2) in $\mathbb{N}^2 \setminus \{(s_1, s_2), (s_2, s_1)\}$ such that $\langle \phi_{t_2}, B\phi_{t_1} \rangle \neq 0$.

Proposition 1.6. Let (A, B, U, Φ) satisfy Assumption 1.1 and assume that if $j \neq k$ and $\lambda_j = \lambda_k$ then $\langle \phi_j, B\phi_k \rangle = 0$ (no internal coupling). If (A, B, U, Φ) admits a non-resonant connectedness chain then the Lie–Galerkin Condition holds.

As a consequence, the Lie–Galerkin Condition is satisfied by a generic single-input bilinear Schrödinger equation, see [104, 111, 121].

1.3.2 Results

The Lie–Galerkin Condition is a generic sufficient condition for approximate controllability.

Theorem 1.7 (Theorem 2.6 in [3]). Let $(A, B_1, \dots, B_p, \Phi, U)$ satisfy Assumption 1.1. If the Lie–Galerkin Condition holds then system (1.2) is approximately controllable by means of piecewise constant controls.

Moreover, approximate controllability with very regular controls holds.

Theorem 1.8 (Theorem 2 in [21]). *Let $(A, B_1, \dots, B_p, \Phi, U)$ satisfy Assumption 1.1 and assume that the set $\mathcal{U}$ of C^∞ functions with values in U is a class of admissible controls. If the Lie–Galerkin Condition holds then system (1.2) is approximately controllable by means of controls in $\mathcal{U}$.*

Actually the Lie–Galerkin Condition goes beyond approximate controllability and implies a stronger controllability property: exact controllability in projections. The main assumption is the continuity of the input-output mapping needed to apply a topological degree argument on the Galerkin approximations. More precisely we assume that the input-output mapping is continuous in the sense that

(A4) if $(u_n)_{n \in \mathbb{N}} \subset \mathcal{U}$ and $u \in \mathcal{U}$ are such that $u_n \rightarrow u$ in $L^1([0, T])$ as $n \rightarrow \infty$ then, for every $\phi \in \mathcal{H}$, $\Upsilon_t^{u_n} \phi$ tends to $\Upsilon_t^u \phi$ in $\mathcal{H}$ uniformly with respect to $t \in [0, T]$ as $n \rightarrow \infty$.

Theorem 1.9 (Theorem 3 in [21]). *Let $(A, B_1, \dots, B_p, \Phi, U)$ satisfy Assumption 1.1 and let $\mathcal{U}$ be the set of piecewise constant functions with values in U . Assume that the Lie–Galerkin Condition holds and that the input-output mapping is continuous as in (A4). Then for every $n \in \mathbb{N}$, for every $\varepsilon > 0$, for every initial condition $\psi_0 \in \mathcal{H}$ with $\|\psi_0\| = 1$, and every final condition $\psi_1 \in \mathcal{H}$ such that $\|\psi_1\| = 1$ and $\|\Pi_n(\psi_1)\| < 1$, there exists $u : [0, T] \rightarrow U$, $u \in \mathcal{U}$ such that*

$$\Pi_n(\Upsilon_T^u(\psi_0)) = \Pi_n(\psi_1) \quad \text{and} \quad \|\Upsilon_T^u(\psi_0) - \psi_1\| < \varepsilon.$$

Remark 1.10. Theorem 1.9 holds true if we replace $\mathcal{U}$ by the set of C^∞ functions with values in U provided that it represents a class of admissible controls.

Remark 1.11. The hypothesis that the final condition ψ_1 satisfies $\langle \phi_j, \psi_1 \rangle > 0$ for some $j > n$ cannot be removed. As mentioned in Section 1.2.4 one cannot expect exact controllability *tout court* if the control operators $B_1, \dots, B_p$ are too regular. The regularity of the control operators and, as a consequence, of the input-output mapping, is therefore an obstruction for the exact controllability while, on the other hand, continuity of the input-output mapping is an assumption needed for the application of the topological degree methods used in the proof of Theorem 1.9. In this sense the controllability in projections is the strongest general controllability result that one may expect in the framework of bounded control potentials.

Remark 1.12. Theorem 1.7 can be extended to simultaneous controllability. Indeed one can prove that it is possible to steer a set of r initial condition arbitrarily close to a unitary combination of them.

Remark 1.13. Under an stronger version of the Lie–Galerkin condition, called Lie–Galerkin tracking condition, it is possible to approximately track a trajectory up to phases, see Section 2.3 in [3].

1.3.3 Examples and extensions

Example 1.14 (Rotating bipolar molecule). We consider the problem of controlling the orientation of a rigid linear molecule in $\mathbb{R}^3$ by external fields presented in Example 1.6. Approximate

controllability for system (1.7) has been proved in [3]. The continuity of the input-output mapping (A4) is guaranteed by the boundedness of the control operators B_1 , B_2 , and B_3 . Moreover the system verifies the Lie–Galerkin Condition.

Note that, because of its symmetries, equation (1.7) is not controllable with only two of the three controls u_1 , u_2 , u_3 . This prevents the use of general result of control focusing on the single-input case. Moreover, the general theory developed in [61, 111, 2] is based on non-resonance conditions on the spectrum of the drift Hamiltonian. The Laplace–Beltrami operator on S^2 , however, has a severely degenerate spectrum, since the ℓ -th eigenvalue $-i\ell(\ell + 1)$ has multiplicity $2\ell + 1$. A perturbation technique has been proposed in [61], in order to overcome resonance relations in the spectrum of the drift. The technique was applied to the case of the orientation of a molecule confined in a plane driven by one control. The planar case is already technically challenging and a generalization of the same technique to the case of three controls in 3D is, in principle, not possible.

Remark 1.15. The infinite potential well of Example 1.5 is another example of system with resonant spectrum for which the Lie–Galerkin Condition and the continuity hypothesis of Theorem 1.9 hold (see Section 2.5 in [3]).

Example 1.16 (The Morse potential with dipolar interaction [5]). The Lie–Galerkin method can be applied also to systems not verifying Assumption 1.1. In particular this geometric methods can be used for the control of systems with continuous part of the spectrum. Consider the dynamics of a molecular bond, such as a $H - H$ bond in a dihydrogen molecule. In this case $\mathcal{H} = L^2((0, \infty), \mathbb{C})$ and $A = -i(-\Delta + V)$ with $V(x) = (1 - e^{-x})^2$. The operator $-\Delta + V$ admits a finite number of simple eigenvalues

$$\lambda_j = h_0 \left(j + \frac{1}{2} \right) - \frac{[h_0 (j + \frac{1}{2})]^2}{4D_e},$$

where $j = 1, \dots, J$ and $h_0 > 0$, D_e , and $J \in \mathbb{N}$ are constants accounting for the physics of the problem. The associated eigenfunctions ϕ_j can be expressed in terms of Laguerre polynomials. The control operator is the multiplicative operator $B = -ix$. A direct computation shows that, for every $j \neq k$,

$$\langle \phi_j, x\phi_k \rangle \neq 0. \quad (1.11)$$

In [5] we proved that the system is approximately controllable in the first two states. In other words, for every $\varepsilon > 0$ there exists u piecewise constant and $T > 0$ such that

$$\|\langle \phi_2, \Upsilon_T^u \phi_1 \rangle - \langle \phi_1, \Upsilon_T^u \phi_2 \rangle\| < \varepsilon.$$

The result can be easily extended to a unitary combination of the first $J - 1$ states. In this particular example the spectrum of A splits in a discrete part and a (possibly essential) part. However, for every $u > 0$, $A + uB$ has compact resolvent (and thus discrete spectrum) and every eigenvector of $A + uB$ belongs to the intersection of the domains of A and B . The idea is to use a small periodic non-zero control to perturb the spectrum and, then, control the system with the Lie–Galerkin method. The convergence of the spectrum of $A + uB$ to the spectrum of A is guaranteed by the Min-Max principle [123, Theorem XIII.1].

Remark 1.17. The control given by the Lie–Galerkin technique is, in some case, optimal for the L^1 norm of the control. In Proposition 1.3 in [12] we proved that the Lie–Galerkin technique minimizes the L^1 norm of the control achieving orientation on the even states of a planar rotating molecule. In [12] we also provided general estimates on the variation of the energy $\langle \psi, A\psi \rangle$ of system (1.2) in the bounded case, in terms of the L^p norm of the control for $p \geq 1$. In [11] we extended these estimates to case of relatively bounded control operators, in terms of the total variation of the control.

Example 1.18. In principle, the Lie–Galerkin method is not designed to control in small time: the control time tends to infinity as the accuracy in the approximate controllability tends to zero. However in some case it is possible to adapt the technique to obtain controllability in arbitrarily small time.

Indeed, consider the bilinear control system

$$i \frac{\partial \psi}{\partial t} = -|\Delta|^\alpha \psi + u(t) \cos \theta \psi, \quad \theta \in \Omega, \quad (1.12)$$

where α is a real constant, $\Omega = \mathbb{R}/2\pi$ is the one dimensional torus, $\mathcal{H} = L^2(\Omega, \mathbb{C})$, and Δ is the Laplace-Beltrami operator on Ω . (For the definition of the operator $|\Delta|^\alpha$ see (1.13) below.)

When $\alpha = 1$ (1.12) is a widely used model for a planar rotating molecule (see for instance Section 8 in [2]). For $\alpha \neq 1$ (1.12) is a purely academical model and has no physical meaning.

The self-adjoint operator $-\Delta$ has purely discrete spectrum $\{k^2, k \in \mathbb{N}\}$. All its eigenvalues are double but zero which is simple. The eigenvalue zero is associated with constant functions. The eigenvalue k^2 for $k > 0$ is associated with the eigenspace spanned by the two eigenfunctions $\theta \mapsto \cos(k\theta)/\sqrt{\pi}$ and $\theta \mapsto \sin(k\theta)/\sqrt{\pi}$. The Hilbert space $\mathcal{H} = L^2(\Omega, \mathbb{C})$ splits in two subspaces $\mathcal{H}_e$ and $\mathcal{H}_o$, the spaces of even and odd functions of $\mathcal{H}$, respectively. The spaces $\mathcal{H}_e$ and $\mathcal{H}_o$ are stable under the dynamics of (1.12), hence one cannot expect global controllability on $\mathcal{H}$. Therefore we consider the restriction of (1.12) to the space $\mathcal{H}_o$. The function $\phi_k : \theta \mapsto \sin(k\theta)/\sqrt{\pi}$ is an eigenvector of the skew-adjoint operator $A = i|\Delta|_{\mathcal{H}_o}|^\alpha$ associated with the eigenvalue $ik^{2\alpha}$, $k \in \mathbb{N}$. The family $(\phi_k)_{k \in \mathbb{N}}$ is a Hilbert basis of $\mathcal{H}_o$. Here, B is the restriction to $\mathcal{H}_o$ of the multiplication by $-i \cos(\theta)$ and it is bounded. Then $\langle \phi_j, B\phi_k \rangle \neq 0$ if and only if $|k - j| = 1$. The set $\{(j, k) \in \mathbb{N} \times \mathbb{N} \mid |j - k| = 1\}$ represents a non-resonant connectedness chain (see Definition 1.9). Hence system (1.12) is approximately controllable in the unit sphere of $\mathcal{H}_o$. Even more: if $\alpha > 5/2$, then for every ψ_0, ψ_1 in the Hilbert unit sphere of $\mathcal{H}_o$, for every $\varepsilon > 0$, for every $T > 0$, there exists a piecewise constant function $u : [0, T] \rightarrow \mathbb{R}$ such that $\|\Upsilon_T^u \psi_0 - \psi_1\| < \varepsilon$. In other words the system is controllable in time zero. This is in contrast with the finite-dimensional case in which there is a non-zero minimal control time. The problem of minimal time in the infinite-dimensional setting is completely open. We refer to [37] and [39] for two results in this direction.

1.4 Weakly-coupled systems

We present here a class of systems that can be precisely approached by finite-dimensional Galerkin approximations: the class of *weakly-coupled* systems.

We define for $s > 0$,

$$|A|^s \psi = \sum_{n \in \mathbb{N}} |\lambda_n|^s \langle \phi_n, \psi \rangle \phi_n, \quad (1.13)$$

for every ψ belonging to

$$D(|A|^s) = \left\{ \psi \in \mathcal{H} \mid \sum_{n \in \mathbb{N}} |\lambda_n|^{2s} |\langle \phi_n, \psi \rangle|^2 < +\infty \right\}.$$

For every $\psi \in D(|A|^s)$ we can define the $|A|^s$ -norm (or simply s -norm) of ψ by $\|\psi\|_s = \| |A|^s \psi \|$. If A is the Laplace–Dirichlet operator on some bounded domain of $\mathbb{R}^n$ then the s -norm is equivalent to the H^{2s} -norm on $D(|A|^s)$.

Assumption 1.2. The tuple $(A, B_1, \dots, B_p, U, \Phi)$ satisfies Assumption 1.1 and the following:

(A5) the operator $i(A + u_1 B_1 + \dots + u_p B_p)$ is bounded from below for every $u \in U$;

(A6) the sequence $(\lambda_k)_{k \in \mathbb{N}}$ is non-increasing and unbounded.

Definition 1.19 ([4]). Let $(A, B_1, \dots, B_p, U, \Phi)$ satisfy Assumption 1.2 and let $s > 0$. Then $(A, B_1, \dots, B_p)$ is s -weakly-coupled if $D(|A + u_1 B_1 + \dots + u_p B_p|^{s/2}) = D(|A|^{s/2})$ for every $u \in U$ and there exists C such that

$$|\Re \langle |A|^s \psi, B_j \psi \rangle| \leq C |\langle |A|^s \psi, \psi \rangle|,$$

for every $j = 1, \dots, p$, $\psi \in D(|A|^s)$.

The terminology *weak-coupling* refers to the smallness of the control operators $B_1, \dots, B_p$ in the scale of A . In other words the effect of $B_1, \dots, B_p$ on the spectral properties of A is small enough to have a weak coupling effect on the Galerkin approximations associated with eigenvectors of A or the boundedness in the s -norm of A of the evolution (see Proposition 1.11 below).

Weak-coupling is satisfied by the majority of physical systems where the operators are very regular.

Proposition 1.10 (Proposition 8 in [4]). Let k be a positive integer, Ω be a compact Riemannian manifold, $V, W : \Omega \rightarrow \mathbb{R}$ be $C^{2k}(\Omega, \mathbb{R})$. Define $A = -i(\Delta + V) : D(A) \rightarrow L^2(\Omega, \mathbb{C})$ and $B = iW : L^2(\Omega, \mathbb{C}) \rightarrow L^2(\Omega, \mathbb{C})$. Then (A, B) is k -weakly-coupled.

A first property of the propagator of a weakly-coupled system is given by the following proposition.

Proposition 1.11 (Proposition 2 in [4]). Let k be positive, $(A, B_1, \dots, B_p, U, \Phi)$ satisfy Assumption 1.2 and be k -weakly-coupled. Then, for every $\psi_0 \in D(|A|^{k/2})$, $K > 0$, $T \geq 0$, and piecewise constant function $u = (u_1, \dots, u_p)$ for which $\sum_{l=1}^p \|u_l\|_{L^1} \leq K$, one has

$$\|\Upsilon_T^u(\psi_0)\|_{k/2} \leq e^{cK} \|\psi_0\|_{k/2}, \quad (1.14)$$

for $c = c(k, A, B_1, \dots, B_p)$ not depending on u and T .

The estimate (1.14) gives a bound on the variation of energy of the controlled system as a function of the L^1 -norm of the control. Finer estimates on the growth of Sobolev norms for linear Schrödinger equation with time dependent Hamiltonian can be found, for instance, in [54]. However the bound in (1.14) does not depend on time nor on the control. This uniformity is crucial in the proof of Theorem 1.12 below.

1.4.1 The Good Galerkin Approximation

One of the most important consequences of the weak-coupling assumption is that, even though the coupling action of the operators $B_1, \dots, B_p$ may give rise to intricate dynamics, this action is weak enough to allow approximations by the dynamics of the truncations. The theorem below states that an infinite-dimensional system can be replaced with arbitrary precision by a suitable Galerkin approximation, called *Good Galerkin Approximation*.

Let

$$\dot{x} = (A^{(N)} + u_1 B_1^{(N)} + \dots + u_p B_p^{(N)})x, \quad x \in \text{span}\{\phi_1, \dots, \phi_N\}, \quad (1.15)$$

be the Galerkin approximation of order $N \in \mathbb{N}$ of system (1.2). We denote by $\Upsilon_t^{(N),u}$ the propagator of (1.15).

Theorem 1.12 (Theorem 4 in [4]). *Let $k, s \in \mathbb{R}$ with $0 \leq s < k$. Let $(A, B_1, \dots, B_p)$ satisfy Assumption 1.2 and be k -weakly-coupled. Assume that there exist $d > 0$ and $0 \leq r < k$ such that $\|B_l \psi\| \leq d \|\psi\|_{r/2}$ for every ψ in $D(|A|^{r/2})$ and l in $\{1, \dots, p\}$. Then for every $\varepsilon > 0$, $K \geq 0$, and ψ_0 in $D(|A|^{k/2})$ there exists $N \in \mathbb{N}$ such that for every piecewise constant function $u = (u_1, \dots, u_p)$*

$$\sum_{l=1}^p \|u_l\|_{L^1} < K \implies \|\Upsilon_t^u(\psi_0) - \Upsilon_t^{(N),u}(\pi_N \psi_0)\|_{s/2} < \varepsilon,$$

for every $t \geq 0$.

As mentioned in Section 1.2.4 the regularity of B with respect to A represents an obstacle to exact controllability. This is the case of weakly-coupled systems as showed in Proposition 1.11. On the other hand, however, weakly-coupled systems can be controlled approximately in strong Sobolev norms (Proposition 5 in [4] and Corollary 2.13 in [3]).

The notion of Good Galerkin Approximation and Theorem 1.12 can be extended to general BV controls (see [8] and [7]). In [10] we extend the notion of weak-coupling in order to drop the assumptions on the spectrum of A and include examples admitting a continuous spectrum.

Remark 1.20. The quantum harmonic oscillator in Example 1.4 is not controllable (in any reasonable sense) as proved in [106]. However, the Galerkin approximations of (1.5) of every order are exactly controllable [130]. Moreover the system is k -weakly-coupled for every k (see Section IV.E of [4]) and Theorem 1.12 ensures that any trajectory of the infinite-dimensional system is a uniform limit of trajectories of its Galerkin approximations. This is not a contradiction, since the infinite-dimensional system cannot track, in general, every trajectory of its Galerkin approximations. In particular, there is no reason for which the infinite-dimensional system

could track a sequence of trajectories of its Galerkin approximations associated with controls with L^1 norm tending to infinity. As a matter of fact, if one wants to steer a solution of the Galerkin approximation of order N of (1.5) from a given state to an ε -neighbourhood of a given target, the L^1 norm of the control blows up as N tends to infinity.

1.4.2 A Bang-Bang theorem for quantum control

The existence of a Good Galerkin Approximation allows to extend finite-dimensional control techniques to the infinite-dimensional setting.

Proposition 1.13 (Lemma 6 in [6]). *Let k be positive and let $(A, B, \Phi, [a, b])$ satisfy Assumption 1.2 and be k -weakly-coupled. Let $T > 0$, $u^* : [0, T] \rightarrow [a, b]$ be a piecewise constant function, and ψ_0 be in $\mathcal{H}$. Then, for every $\varepsilon > 0$, there exists a piecewise constant control $u_\varepsilon : [0, T_\varepsilon] \rightarrow \{a, 0, b\}$ such that $\|\Upsilon_{T_\varepsilon}^{u_\varepsilon}(\psi_0) - \Upsilon_T^{u^*}(\psi_0)\| < \varepsilon$, and $\|u_\varepsilon\|_{L^1} \leq \|u^*\|_{L^1}$. Moreover, if u^* is positive, then u_ε may be chosen with values in $\{0, b\}$.*

The idea of Proposition 1.13 is that an approximately controllable weakly-coupled system can be controlled by means of a control taking only three values (two in the positive case). Proposition 1.13 is classical in finite dimension and it can be obtained, for instance, with Lie groups techniques, see [127]. The idea for extend it to the infinite dimension is to take the approaching sequence $(u_n)_n$ uniformly bounded in $L^1(\mathbb{R}, \mathbb{R})$ in order to exploit the weak-coupling and, in particular, Theorem 1.12.

Remark 1.21. A consequence of Proposition 1.13 is an approximate controllability result (Theorem 1 in [6]) of the bilinear Schrödinger equation with a polarizability term of the form $\dot{\psi} = (A + u(t)B_1 + u(t)^2B_2)\psi$, (see also [107]).

1.5 Perspectives

Quantum control represents a research field very rich in complex challenges coming both from the mathematical framework and from the physical applications. The interest of quantum control lies precisely in the fact that the two aspects, theory and applications, nourish each other.

A mathematically challenging and physically relevant open problem is the orientation of a non-symmetric molecule, which is modeled by the Schrödinger equation on $\text{SO}(3)$, or the case of a molecule presenting a “thin” degree of freedom. In the last case the drift Hamiltonian contains a sub-Laplacian and it would be interesting to study both the free evolution (many questions are still open for the hypoelliptic Schrödinger equation) and the controllability. In order to tackle these challenges we could exploit the Lie–Galerkin technique and combine it with the generalized Fourier transform, which is a helpful tool to investigate the spectral properties of the Schrödinger operator when the state space is a Lie group. Indeed, one of the main advantages of the Lie–Galerkin technique is that the results apply to very general systems and, in particular, to those with strong spectral degeneracies and geometrically non-trivial state space.

Exact controllability represents a very hard open mathematical challenge. The problem could be attacked by two sides: on one hand by studying the regularity of the propagators of the bilinear Schrödinger equation and, on the other hand, by improving the results of approximate controllability. For what concerns the regularity of the propagators the problem is to find the good compromise on the hypotheses of the system in order to enlarge the class of admissible controls without losing exact controllability. A fundamental tool in this context is represented by the energy estimates. Preliminary results in this direction are in [10], where we extend the definition of weakly-coupled systems and the existence of Good Galerkin Approximations to a large class of controls and to general dissipative bilinear systems.

As concerns controllability, exact controllability in projections (Theorem 1.9) is the strongest result of controllability in the case of regular propagators (Remark 1.11). However, being very general, the result is far from optimal in terms of controllability time. The minimization of the time needed to achieve a transfer of population towards a given target is also a major issue in all applications to physics of quantum control. Minimal time for controllability is a measure of the feasibility of the control strategy from the point of view of physical constraints. The existence of a minimal controllability time is an indicator of the non-controllability of the system [37].

Improving the available time estimates for approximate controllability is, therefore, of paramount interest. The idea is to use Good Galerkin Approximations to obtain sufficient conditions and necessary conditions for approximate controllability in arbitrarily small time. In the best case, one could obtain a sufficient and necessary condition for approximate controllability in arbitrary small time. Otherwise, standard finite-dimensional techniques and *a priori* energy estimates could provide separately sufficient conditions and necessary conditions. It is also interesting to design further examples for which approximate controllability in arbitrary small time holds, as in Example 1.18.

A long term project could be the study of open quantum systems, i.e., of systems for which the norm of the state is not preserved during the evolution due to the interrelation with the environment. Open systems are the natural complement for the analysis of rotational models for molecules. Models for dense gas samples in which molecular collisions play an important role are based on Lindblad equations. The idea is to extend to open systems the notion of weakly-coupled systems and the error estimates between the evolution of the infinite-dimensional closed quantum system and its Galerkin approximations.

Finally, wave equations, both bilinear and non-linear represent a class of systems admitting a sequence of Galerkin approximations. The extension of methods of finite-dimensional geometric control theory to this class of equations is possible in principle.

Complex systems

2.1 Introduction

A complex system is a dynamical system composed of several interacting components, or agents. Systems of this kind are sometimes called *multiagent systems* and are usually represented as a network where the nodes account for the agents and the links for their interactions. A fascinating feature of large groups of autonomous agents is their ability to form organized global patterns even when individual agents interact only at a local scale. This is usually referred to as *self-organization* and the resulting phenomenon is called *emergent behavior*. The observation of this features of complex systems dates back to the 19th century. The term *emergent*, indeed, was coined by the philosopher G. H. Lewes who, in 1875, wrote:

Every resultant is either a sum or a difference of the cooperant forces [...] The emergent is unlike its components insofar as these are incommensurable, and it cannot be reduced to their sum or their difference.

Since then, multiagent systems attracted the attention of researchers from many disciplines. The problem of describing emergent behaviors in a group of autonomous agents has been the object of study in a number of situations ranging from linguistics to distributed computing and from physics to animal behavior. Examples of networks of agents are, indeed, ubiquitous: biological networks (genetic regulation, ecosystems), technological networks (internet), economical networks (production, distribution, and financial networks) and social networks (collaboration networks, crowd movement), etc. The challenge is then to describe complex behaviors by assigning only simple and local interaction rules.

A common feature to many complex systems is that they represent a natural example of dissipative systems, that is, they are dynamical systems, finite- or infinite-dimensional, admitting an energy functional (e.g. a Lyapunov function) which is not increasing along the trajectories of the system. The intrinsic relation between dissipative systems and self-organization has been pointed out in thermodynamics by the seminal works [114, 120] introducing the scientific community to the analysis of self-organized dynamics and, more in general, to complex systems.

From the mathematical viewpoint, models of complex systems can be classified as first-order models and second-order models. In first-order models, the variables of interest are called *opinions*, even though such models can describe a wide range of attributes such as positions, market shares or wealth. The opinion of each agent is affected by neighboring agents' opinions in the state space. On the other hand, in second-order models, the variables of interest are

velocities, obtained as the time derivatives of the positions. Each agent's velocity is affected by the velocities of agents whose positions are close in the state-space.

Another level of classification of the models is given by the scale. We can identify two scales depending on the parameters and variables used for modeling. At *microscopic scale* a finite number of interacting agents are described by their position and, possibly, their velocity which are time-dependent variables. The dynamics is given by a system of ordinary differential equations. In the *macroscopic scale* the state of the system is described by averaged quantities, such as, density, linear momentum, and kinetic energy. This description is useful when dealing with a large number of agents. The time evolution of these quantities is governed by systems of partial differential equations. We also refer to this kind of models as infinite-dimensional, in contrast with the finite-dimensional description at the microscopic scale.

We focus only on first-order and second-order dissipative models, as presented in Sections 2.1.1 and 2.1.2 respectively. For a complete review we refer to [1] in which we present a survey on first-order and second-order models depending on the interaction network and on the state space. The survey [110] presents a unified approach to describe first-order and second-order models in finite-dimension as well as an introduction to the analysis of the models at macroscopic scale. We refer, moreover, to [42] for a survey of models in traffic flows and crowd dynamics and a detailed analysis on the differences between the scales and to [41, 44] for a general analysis of modeling crowd dynamics and complex systems.

2.1.1 First-order models

First-order models are also referred to as *consensus* or *opinion formation* models. A first formulation of linear opinion formation models can be traced back to French's research on social influence [78]. More recently, non-linear models were introduced and analyzed in [99, 98, 73, 85].

The general model describes N agents, each with a vector of position $x_i \in \mathbb{R}^d$, interacting with each other according to

$$\dot{x}_i = \sum_{j \neq i} a_{ij}(x_j - x_i), \quad \text{for } i = 1, \dots, N, \quad (2.1)$$

for some coefficients $a_{ij} \geq 0$, possibly depending on $x = (x_1, \dots, x_N)$. One of the most influential models in opinion formation is the Bounded Confidence Model by Hegselmann and Krause [85]. The main feature of this model is that the interaction is zero when the distance between two opinions is larger than a certain threshold, namely

$$a_{ij} = \begin{cases} 1 & \text{if } |x_i - x_j| \leq r, \\ 0 & \text{otherwise.} \end{cases}$$

This model captures phenomena such as formation of consensus and emergence of clustering, depending on the size of the interaction regions and the density of agents in the domain. For almost every initial configuration, opinions converge asymptotically to clusters [50]. And if the interaction is strong enough (i.e. r is large enough) or all agents are sufficiently close to

consensus then consensus emerges, i.e. the solution tends to $\{x_1 = \dots = x_N\}$. Consensus is therefore a locally attractive equilibrium. Even more, the system is dissipative, meaning that the function $V(x) = \max_{i,j} \|x_i - x_j\|^2$ is a proper Lyapunov function for the system for any adjacency matrix with coefficients a_{ij} , $i, j = 1, \dots, N$. A wide variety of models have been developed by changing the confidence region and the coefficients a_{ij} , see for instance [91, 110].

When the number of agents is very large it is convenient to leave the microscopic viewpoint where each agent is considered individually, and consider instead the mean-field limit, which provides a kinetic description of the system. This approach consists of approximating the influence of all agents on any given individual by an averaged effect. At macroscopic scale the kinetic description of (2.1) when $a_{ij}(x) = \phi(x_i - x_j)$ for some Lipschitz function ϕ reads

$$\partial_t \mu + \nabla \cdot (f[\mu] \mu) = 0, \quad \mu(0) = \mu_0 \quad (2.2)$$

where μ_0 is the initial distribution of agents' opinion, μ is a time evolving probability measure and

$$f[\mu](x) = \int \phi(y - x)(y - x) d\mu(y). \quad (2.3)$$

The kinetic version of the Hegselmann–Krause model has been first studied in [59] for discrete-time dynamics. Existence of solutions has been proved in [45]. If $\phi(\cdot)$ is non-vanishing, then the system converges unconditionally to consensus, meaning that for every μ_0 the solution $\mu(t)$ converges asymptotically to a Dirac mass. If $\phi(\cdot)$ is compactly supported however, then the large time behavior of the dynamics is not yet completely understood and a precise description of the asymptotic dynamics is, in general, a hard task. As in the finite-dimensional analogue, generically the solution $\mu(t)$ converges to a finite sum of Dirac deltas, representing clusters of opinion, but sufficient conditions for global consensus are still unknown.

2.1.2 Second-order models

Second-order models are commonly applied to animal groups to study coordinated collective behaviors [66, 97, 137, 82, 67, 103], for example fishes [89, 116] or birds [70, 32]. Some models have been designed to include simple interaction rules like attraction, short-distance repulsion and mimetic orientation or alignment. Agreement of all agents in the velocity variable is referred to as *alignment* or *flocking*. The prototypical second-order model for the interaction of N agents is the *Cucker-Smale model* (CS) [70]

$$\begin{cases} \dot{x}_i(t) &= v_i(t) \\ \dot{v}_i(t) &= \frac{1}{N} \sum_{j=1}^N a(\|x_j(t) - x_i(t)\|)(v_j(t) - v_i(t)), \end{cases} \quad i = 1, \dots, N, \quad (2.4)$$

where $x_i \in \mathbb{R}^d$, $v_i \in \mathbb{R}^d$, and $a \in C^1([0, +\infty))$ is a non-increasing positive function called *interaction potential* and accounting for the rate of communication. In the classical CS model, we have $a(s) = (1 + s^2)^{-\beta}$, with $\beta > 0$. Here, x_i is the main state of agent i , and v_i is its consensus parameter. This model was initially introduced to describe the formation and

evolution of languages, and was then also used to describe the flocking of a swarm of birds [70] or spacecraft formation [117] and boasts several extensions, see for instance [132, 25, 46].

In the Cucker–Smale system the mean velocity $\bar{v} = \frac{1}{N} \sum_{i=1}^N v_i(t)$ is an invariant of the motion and the velocity variance

$$V(t) = \frac{1}{2N^2} \sum_{i,j=1}^N \|v_i(t) - v_j(t)\|^2 = \frac{1}{N} \sum_{i=1}^N \|v_i(t) - \bar{v}\|^2, \quad (2.5)$$

is a Lyapunov function since

$$\dot{V}(t) = -\frac{1}{N} \sum_{i,j=1}^N a(\|x_j(t) - x_i(t)\|) \|v_i(t) - v_j(t)\|^2 \leq 0.$$

The manifold $\{(x, v) \in (\mathbb{R}^d)^N \times (\mathbb{R}^d)^N \mid v_1 = \dots = v_N\}$, called *consensus* or *flocking manifold*, is locally attractive. An estimate of its basin of attraction, called *flocking region* in Figure 2.1, is given in terms of the velocity variance and the spatial variance $X(t) = \frac{1}{2N^2} \sum_{i,j=1}^N \|x_i(t) - x_j(t)\|^2$ by the following result.

Proposition 2.1 (Theorem 3.1 in [83]). *Let $(x_0, v_0) \in (\mathbb{R}^d)^N \times (\mathbb{R}^d)^N$ be such that*

$$\sqrt{V(0)} \leq \int_{\sqrt{X(0)}}^{+\infty} a(\sqrt{2Nr}) dr.$$

Then the solution with initial data (x_0, v_0) tends to flocking.

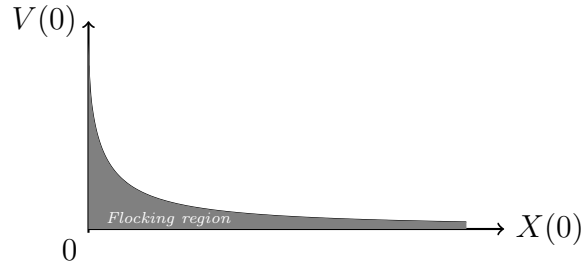


Figure 2.1: Flocking region $\sqrt{V(0)} \leq \frac{1}{\sqrt{2N}} (\frac{\pi}{2} - \arctan(\sqrt{X(0)}))$ corresponding to the CS model (2.4) with parameter $\beta = 1$, see Proposition 2.1.

Even if the system is dissipative there are initial conditions outside the flocking region not converging to flocking as the following example shows.

Example 2.1. Consider the alignment model (2.4) in the case of $N = 2$ agents moving in $\mathbb{R}$ with position and velocity at time t , $(x_1(t), v_1(t))$ and $(x_2(t), v_2(t))$ respectively and with interaction potential $a(x) = 2/(1 + x^2)$. Let $x(t) = x_1(t) - x_2(t)$ be the relative state and $v(t) = v_1(t) - v_2(t)$ be the relative velocity. Then (2.4) reads $\dot{x} = v, \dot{v} = -v/(1 + x^2)$. The

solution with initial conditions $x(0) = x_0$ and $v(0) = v_0 > 0$ is, by direct integration, $v(t) - v_0 = -\arctan x(t) + \arctan x_0$. Whenever the initial conditions satisfy $|\arctan x_0 + v_0| > \pi/2$, which implies $|\arctan x_0 + v_0| \geq \pi/2 + \varepsilon$ for some $\varepsilon > 0$, the velocity $v(t)$ remains bounded away from 0 for every time, since

$$|v(t)| = |-\arctan x(t) + \arctan x_0 + v_0| \geq |-\arctan x(t) + \pi/2 + \varepsilon| > \varepsilon,$$

for every $t > 0$. In other words, the system does not tend to flocking.

At macroscopic scale, by taking the mean-field limit in system (2.4), we obtain the transport PDE with non-local term, called kinetic Cucker–Smale system

$$\partial_t \mu + \operatorname{div}_{(x,v)} (V[\mu] \mu) = 0, \quad (2.6)$$

with the velocity field

$$V[\mu] = \begin{pmatrix} v \\ \xi[\mu] \end{pmatrix},$$

where $\mu(t)$ is a probability measure on $\mathbb{R}^d \times \mathbb{R}^d$ (if $\mu(t, x, v) = f(t, x, v) dx dv$, then f is the density of the particles), and $\xi[\mu]$ is the *interaction kernel*, given by

$$\xi[\mu](x, v) = \int_{\mathbb{R}^d \times \mathbb{R}^d} a(\|x - y\|)(w - v) d\mu(y, w).$$

The solution to the kinetic equation (2.6) is formally

$$\mu(t) = \Phi(t) \# \mu_0,$$

that is, the pushforward under the *particle flow* $\Phi(t)$ generated by $V[\mu(t)]$ of the initial measure μ_0 . The derivation and the asymptotic analysis of (2.6) is given in [84, 60]. For a general result on the derivation of kinetic models we refer to [58].

2.1.3 External control

Despite the existence of a global Lyapunov function both first-order and second-order systems present solutions not converging to the global consensus. When consensus is not achieved by self-organization, it is then natural to wonder whether it is possible to steer the group to consensus by means of an external action. This is the problem of the *organization via intervention*, in contrast with the notion of self-organization. A large number of applications involve control of robotic networks or autonomous vehicles [57] or to guide the agents in a specific direction, as done for the migration of animal groups in [103].

The first control theoretical studies applied to complex systems date back to the the first 2000s. In this framework, consensus in multiagents systems has been seen as a distributed control problem [92, 108, 109, 115]. The problem is to find a feedback control, or *consensus protocol*, $u(x) = (u_1(x), \dots, u_N(x))$ assigning dynamics to the i -th agent $\dot{x}_i = u_i(x)$ for $i = 1, \dots, N$ such that every solution tends to a consensus configuration $x_1 = \dots = x_N$. These pioneering results however only focus on systems with trivial free dynamics, i.e. there is no evolution when $u_i = 0$ for $i = 1, \dots, N$.

Distributed stabilization

Here we set the problem in the more general framework of control-affine systems on $\mathbb{R}^n$ when the free evolution of the system is dissipative, i.e. when there exists a non-strict Lyapunov function, as in the above-mentioned cases of sections 2.1.1 and 2.1.2. Since consensus represents in some sense a steady configuration for the system, enforcing self-organization can be seen as an asymptotic stabilization problem, which is classical in control theory and usually relies on Lyapunov design (see [136]). For instance, the well-known Jurdjevic–Quinn Theorem [94] gives an explicit expression for a continuous global stabilizer.

Let n and m be positive integers, let f and g_i , $i = 1, \dots, m$ be smooth vector fields defined on $\mathbb{R}^n$, and let U be a convex subset of $\mathbb{R}^m$ containing a neighborhood of the origin. We consider the control-affine system in $\mathbb{R}^n$

$$\dot{x}(t) = f(x(t)) + \sum_{i=1}^m u_i(t)g_i(x(t)), \quad (2.7)$$

where the control $u = (u_1, \dots, u_m)$ takes its values in U . We assume the uncontrolled system (i.e., with $u \equiv 0$) to be *dissipative*, meaning that there exists a smooth function $V : \mathbb{R}^n \rightarrow \mathbb{R}$ such that:

- V is radially unbounded (or proper), i.e., $V^{-1}((-\infty, \ell])$ is compact for every $\ell \in \mathbb{R}$;
- $L_f V(x) \leq 0$ for every $x \in \mathbb{R}^n$.

Here $L_f V(x) = \langle \nabla V(x), f(x) \rangle$ denotes the *Lie derivative* of V along f .

According to the well-known Jurdjevic–Quinn theorem (see [94]), if we assume that $f(0) = 0$ and that

$$\{x \in \mathbb{R}^n \mid L_f V(x) = 0 \text{ and } L_f^k L_{g_i} V(x) = 0, \text{ for } i = 1, \dots, m, k \in \mathbb{N}\} = \{0\},$$

then the smooth feedback defined by

$$u(x) = -(L_{g_1} V(x), L_{g_2} V(x), \dots, L_{g_m} V(x)), \quad (2.8)$$

globally asymptotically stabilizes system (2.7) to 0. A more general version of this result provides convergence to some invariant set, described in terms of the Lie derivatives of V . The convergence is established thanks to LaSalle invariance principle. The importance of this result lies in the fact that it can be easily applied to first-order and second-order dissipative models to design a stabilizing feedback.

Sparse stabilization

Feedback stabilizers designed by means of classical methods, such as the Jurdjevic–Quinn control in (2.8), may have, in general, several non-zero components. In terms of multiagent dynamics, acting on every agent may be unfeasible in practice, particularly when dealing with large groups of agents. It is therefore natural to seek controls achieving the same goal with

less active components. This is the case, for instance, when we look for a single leader to act on a whole crowd [28, 53, 27, 76], [23, 22] or, more generally, when feasible control strategies are required to focus on a small number of agents at each time [15, 14]. We address here the question: is it possible to design a feedback strategy in which only a minimal number of components are active at each instant of time? This question is inspired by the works [15, 14] introducing the notion of *sparse control*.

The notion of “sparsity” arises naturally in optimal control of multiagent systems. Indeed consider the following simple linear consensus model with an external control

$$\dot{x}_i = \sum_{j=1}^N a_{ij}(x_j - x_i) + u_i, \quad i = 1, \dots, N,$$

where $x_i \in \mathbb{R}^d$, $a_{ij} \in \mathbb{R}$ and the control $u = (u_1, \dots, u_N)$ verifies the constraint

$$\sum_{i=1}^N \|u_i\| \leq M,$$

for some given $M > 0$. Consider the problem of steering the system to the consensus $x_1 = \dots = x_N$ in minimal time T . Then the Pontryagin Maximum Principle ensures the existence of a absolutely continuous non-trivial vector function $(p_1(t), \dots, p_N(t))$ satisfying the adjoint equation

$$\dot{p}_i = \sum_{j=1}^N a_{ij}(p_i - p_j), \quad i = 1, \dots, N,$$

with final constraint $\sum_i p_i(T) = 0$. The minimality condition for the time-optimal control problem reads

$$\min \sum_{i=1}^N \langle p_i(t), u_i(t) \rangle,$$

so that the optimal control is, if $\|p_i(t)\| > \|p_j(t)\|$ for every $j \neq i$,

$$u_i = -M \frac{p_i(t)}{\|p_i(t)\|}, \quad \text{and } u_j = 0 \text{ for } j \neq i.$$

In particular, the time optimal control is sparse except when the index for which $\|p_i(t)\|$ is maximal is not unique. However, in the generic case in which all interaction coefficients a_{ij} are pairwise distinct then the time optimal control is sparse for almost every $t \in [0, T]$.

2.1.4 Contributions

In [15] we introduced the notion of sparse control, while seeking an optimal control for the Cucker–Smale system (2.4). We proved that it is possible to steer the system to the equilibrium by acting only on a single agent at each instant of time. The idea is based on a geometric constraint of the ℓ^1 -type on the control in order to enforce sparsity in the vector of control

functions. The result then extends to general alignment models in [14] in which we prove that the sparsest control is also the “best” one.

In parallel, we also considered the problem of control via leader, that is acting on a single component with “strong” interactions. This kind of problem can be seen as a sparse controllability problem in the limit case in which the set of controllable components (or agents) is reduced to a single element, the leader. In this framework we studied local controllability, global stabilization and optimal control for the Hegselmann–Krause model (2.1) in [23], and for a non-linear model of network evolution, the Heider balance model, in [22].

Recently, we extended the results of sparse stabilization to the class of control-affine systems with a non-strict Lyapunov function [20]. This class includes the above mentioned models [19] as well as many multiagent models. Thanks to a mean-field limit procedure we adapted these result to transport PDEs with non-local terms [18] providing a control method for kinetic multiagent models.

When dealing with sparse controls of large groups chattering phenomena naturally arise. In [16] we studied the problem of chattering in general non-linear optimal control problems. We defined a sequence of suboptimal problems with regular solutions and we analysed the rate of convergence to the optimal ones.

We treated also the modeling problem and we implemented models of opinion formations on manifolds [17] as well as pedestrian models with psychological insight [13].

2.1.5 Contents

In Section 2.2 we present general results in sparse control for dissipative systems (Section 2.2.1) and for transport equations with non-local term (Section 2.2.2). These results are applied, in Section 2.3, to the sparse stabilization of second-order alignment models (Section 2.3.1) and to the control via leader of social models (Section 2.3.2). We also present results of local controllability and optimal control in these particular examples.

In Section 2.4 we treat the problem of chattering phenomena in general non-linear optimal control problems.

In Section 2.5 we address briefly the modeling problem. A non-linear model of opinion formation on the sphere is presented in Section 2.5.1. In Section 2.5.2 we mention the results of two experiments designed to determine the psychological bias involved in human locomotion, useful for the definition of a new class of cognitive-mathematical models.

We conclude, in Section 2.6, with a description of possible developments.

2.2 General results in sparse control

2.2.1 Sparse stabilization of dissipative systems

Here we consider the problem of finding sparse stabilizers for a dissipative control-affine system of the form (2.7). We provide three different control strategies to achieve stabilization by using a sparse Jurdjevic–Quinn controller, mimicking the form (2.8), namely a sparse controls of the form $u_i(x) = -L_{g_i}V(x)$ for some $i \in \{1, \dots, m\}$, while $u_j(x) = 0$ for $j \neq i$. The key aspect

is to determine the strategy to switch from one active component of the control to another. Indeed, discontinuity issues in the definition of sparse stabilizers arise naturally, as shown for instance in [15, 14, 19]. (See also Section 2.4 below for general the case of non-linear control problems.) Here we develop three different approaches to deal with discontinuous feedbacks, each of them leading to a different kind of sparse stabilizer: a time-varying periodic feedback, a sampled feedback, and a hybrid feedback.

The absence of continuous feedback stabilizers is a classical matter in control (as in [56]) and a classical approach [135, 128] is to use time-varying periodic feedback controls (see also [65, 125] or [64, section 11.2]). In this spirit, we consider a first strategy, as follows.

Strategy 1: Sparse time-varying periodic feedback.

Fix the sampling time $\tau > 0$ and the final control time $T > 0$. For the initial state $x_0 \in \mathbb{R}^n$, consider the unique trajectory $x(t)$ of (2.7) with the time-varying feedback control $u(t, x) = (u_1(t, x), \dots, u_m(t, x))$ defined by

$$u_i(t, x) = -L_{g_i}V(x), \quad \text{and} \quad u_j(t, x) = 0, \quad \text{for every } j \neq i,$$

for every $t \in [(km + i - 1)\tau, (km + i)\tau) \cap [0, T]$ for $k \in \mathbb{N}$ and $i = 1, \dots, m$. With $u(t, x) = 0$ for $t \geq T$.

In our second sampling approach, we discretize the time horizon and we apply a fixed control u_i on each interval. Such a control is chosen with a steepest descent approach, by maximizing the instantaneous decrease of V at the beginning of the sampling interval.

Strategy 2: Sampled sparse feedback.

Consider the sparse feedback $u(x) = (u_1(x), \dots, u_m(x))$ defined at any $x \in \mathbb{R}^n$ by

$$u_i(x) = -L_{g_i}V(x) \quad \text{and} \quad u_j = 0 \quad \text{for } j \neq i, \quad (2.9)$$

where $i \in \{1, \dots, m\}$ is the smallest integer such that

$$|L_{g_i}V(x)| \geq |L_{g_j}V(x)|, \quad \forall j \neq i. \quad (2.10)$$

Fix a sampling time $\tau > 0$. Then consider the sampling solution associated with u and the sampling time τ , namely the solution of

$$\dot{x}(t) = f(x(t)) + \sum_{i=1}^m u_i(x(k\tau))g_i(x(t)), \quad t \in [k\tau, (k+1)\tau],$$

with $k \in \mathbb{N}$.

The notion of stabilization associated with sampling solutions is the stabilization in the sample-and-hold sense (see for instance [62, Section 7]).

Definition 2.2. Let $U \subset \mathbb{R}^m$, let $F : \mathbb{R}^n \times U \rightarrow \mathbb{R}^n$ be continuous and locally Lipschitz in x , uniformly on compact subsets of $\mathbb{R}^n \times U$, with $F(\bar{x}, 0) = 0$. We say that a feedback $u : \mathbb{R}^n \rightarrow U$ stabilizes the system $\dot{x} = F(x, u(x))$ to $\bar{x}$ in the sample-and-hold sense if for every $r > 0$ and $R > 0$ there exists $\tau > 0$ and $T > 0$ depending only on r and R and $C > 0$ depending only on R such that for any $x_0 \in B_R(\bar{x})$ the sampled solution of $\dot{x} = F(x, u(x))$, $x(0) = x_0$, with sampling time τ satisfies $|x(t)| \leq C$ for every $t \geq 0$ and $x(t) \in B_r(\bar{x})$ for every $t \geq T$.

Finally, we consider a hybrid approach based on *hysteresis*: we choose a component of the control u_i maximizing the instantaneous decrease of V . This component is the only active one as long as it satisfies the lower threshold condition $|L_{g_i}V(x)| > (1 - \varepsilon)|L_{g_j}V(x)|$ for any $j \neq i$. When such lower threshold is reached, the control switches to the new control maximizing the instantaneous decrease of V .

Strategy 3: Sparse feedback with hysteresis.

Fix $\varepsilon \in (0, 1)$ and apply the following algorithm to define the trajectory $x(t)$ of the system:

- **Initialize:** $n = 0$ and $t_0 = 0$.
- **While** $t_n < +\infty$ **apply Step** n : At time t_n choose $i = 1, \dots, m$ being the smallest integer such that

$$|L_{g_i}V(x(t_n))| \geq |L_{g_j}V(x(t_n))|, \quad \text{for every } j \neq i. \quad (2.11)$$

- If $|L_{g_i}V(x(t_n))| \geq 2t_n^{-1}$, define the switching time t_{n+1} as the infimum of times $t \in [t_n, +\infty)$ such that the unique solution $y(t)$ of $\dot{y} = f(y) - L_{g_i}V(y)g_i(y)$ with $y(t_n) = x(t_n)$ satisfies

$$|L_{g_i}V(y(t))| \leq t^{-1} \quad \text{or} \quad |L_{g_i}V(y(t))| \leq (1 - \varepsilon)|L_{g_j}V(y(t))|, \quad \text{for some } j \neq i, \quad (2.12)$$

with the convention that $t_{n+1} = +\infty$ if the solution satisfies

$$|L_{g_i}V(y(t))| > t^{-1} \quad \text{and} \quad |L_{g_i}V(y(t))| > (1 - \varepsilon)|L_{g_j}V(y(t))|,$$

for every $j \neq i, t \geq t_n$. Define the control $u(t, x) = (u_1(x), \dots, u_m(x))$, $u_i(x) = -L_{g_i}V(x)$, $u_j(x) = 0$ for $j \neq i$ and the corresponding trajectory $x(t)$ on the interval $[t_n, t_{n+1})$.

- If $|L_{g_i}V(x(t_n))| < 2t_n^{-1}$, define the switching time t_{n+1} as the infimum of times $t \in [t_n, +\infty)$ such that the unique solution $y(t)$ of $\dot{y} = f(y)$ with $y(t_n) = x(t_n)$ satisfies

$$|L_{g_j}V(y(t))| \geq 4t^{-1}, \quad \text{for some } j = 1, \dots, m, \quad (2.13)$$

with the convention that $t_{n+1} = +\infty$ if the solution satisfies

$$|L_{g_j}V(y(t))| < 4t^{-1} \quad \text{for all } j = 1, \dots, m, \quad t \geq t_n.$$

Define the control $u(t, x) = 0$ and the corresponding trajectory $x(t)$ on the interval $[t_n, t_{n+1})$.

- If $t_{n+1} < +\infty$, **pass from Step** n **to Step** $n + 1$.

The three sparse control strategies above asymptotically stabilize a dissipative control-affine system.

Theorem 2.2 (Theorem 1.1 in [20]). Assume that there exists a smooth function $V : \mathbb{R}^n \rightarrow \mathbb{R}$ such that

- V is radially unbounded (or proper), i.e., $V^{-1}((-\infty, \ell])$ is compact for every $\ell \in \mathbb{R}$;
- $L_f V(x) \leq 0$ for every $x \in \mathbb{R}^n$.

Let

$$\mathcal{Z} = \{x \in \mathbb{R}^n \mid L_f V(x) = L_{g_i} V(x) = 0, \text{ for every } i = 1, \dots, m\},$$

and Ω be the largest subset of $\mathcal{Z}$ that is invariant under the flow of $\dot{x} = f(x)$. Then:

- (i) If Ω is locally attractive, then, for each initial state $x_0 \in \mathbb{R}^n$, there exist $\tau_1 > 0$ and $T > 0$ such that, for every $\tau \in (0, \tau_1)$, Strategy 1 with sampling time τ and final control time T asymptotically stabilizes the control system (2.7) to Ω .
- (ii) If $\mathcal{Z} = \{\bar{x}\}$ for some $\bar{x} \in \mathbb{R}^n$, then, for every initial state $x_0 \in \mathbb{R}^n$, there exists $\tau_2 > 0$ such that, for every $\tau \in (0, \tau_2)$, Strategy 2 with sampling time τ asymptotically stabilizes the control system (2.7) to $\bar{x}$, in the sample-and-hold sense.
- (iii) For every $\varepsilon \in (0, 1)$, Strategy 3 asymptotically stabilizes the control system (2.7) to Ω .

Moreover, τ_1 and τ_2 can be chosen uniformly with respect to all initial conditions belonging to an arbitrary compact subset of $\mathbb{R}^n$.

Let us point out the major advantages and drawbacks of each strategy. Strategy 1 requires local attractivity of the set Ω and, in order to design the control, the Lie derivatives of $V(x)$ are required to be computed at $x(t)$ for any t . Strategy 2 applies if the set $\mathcal{Z}$ reduces to a single point, but the evaluation of $V(x(\cdot))$ is only required to be performed at discrete times $n\tau$, for $n \in \mathbb{N}$. The stabilization to $\mathcal{Z}$ is realized in the sample-and-hold sense (Definition 2.2). Notice that whenever the equilibrium is locally attractive (as it is the case for the multiagent models considered in Section 2.3 below) stabilization in the sample-and-hold sense is equivalent to stabilization *tout court*. Strategy 3 stabilizes the system for any parameter ε , and hence it is in general more robust than Strategies 1 and 2; however it is required to evaluate $V(x(\cdot))$ at any t along the trajectories.

First-order and second-order dissipative multiagent models as presented in sections 2.1.1 and 2.1.2, fall within the framework of Theorem 2.2. Hence this result provides three explicit sparse control strategies to steer a system to consensus/alignment, represented by the invariant singular set Ω . In particular, Strategy 2 has been applied to the Cucker–Smale model (2.4) in [15], to general alignment models in [14], to the controllability via leader of the Hegselmann–Krause model (2.1) in [23]. We present these applications in Section 2.3 below. Finally notice that Strategy 2 has been adapted also to a non-dissipative system, the Cucker–Dong model [69], to obtain a local result in [51].

2.2.2 Sparse control of Mean-Field equations

As showed in sections 2.1.1 and 2.1.2 many kinetic equations obtained via mean-field limit are transport equations with non-local velocity of the form

$$\partial_t \mu + \nabla \cdot (f[\mu] \mu) = 0, \tag{2.14}$$

where μ is a probability measure on $\mathbb{R}^d$ representing the density of agents, $\nabla \cdot$ is the divergence operator, and $f[\mu]$ is a vector field depending on the measure accounting for the interactions between agents.

The control problem associated with (2.14) is

$$\partial_t \mu + \nabla \cdot ((f[\mu] + u(t, \cdot)g[\mu])\mu) = 0, \quad \mu(0) = \mu_0, \quad (2.15)$$

where μ_0 is a probability measure on $\mathbb{R}^d$ representing the initial density of agents.

Here, given a measurable subset ω of $\mathbb{R}^d$, we denote by $|\omega|$ its Lebesgue measure and by χ_ω its indicator function, defined almost everywhere by $\chi_\omega(x) = 1$ if $x \in \omega$ and $\chi_\omega(x) = 0$ otherwise.

Definition 2.3. Given $c > 0$, an *admissible control* is

$$u : (t, x) \mapsto \chi_{\omega(t)}(x)v(t, x),$$

such that

- (i) $|\omega(t)| \leq c$;
- (ii) $(t, x) \mapsto v(t, x)$ is Lipschitz in x and piecewise constant in t such that $\|v(t, \cdot)\|_\infty \leq 1$ for almost every $t \geq 0$.

We denote by $\mathcal{U}$ the set of admissible controls.

The rationale for the choice of the control is given by the fact that agents are indistinguishable, so that controls can only be state-dependent and cannot focus on specific agents. The control action is therefore given by a vector field $g[\mu]$, a *control gain* $v(t, x)$, and a *control set* $\omega(t)$ on which the control gain is localized. Notice that the control gain v and the control set ω are both time-dependent while the vector field g is fixed, and it depends on the density μ .

Remark 2.4. Condition (i) in Definition 2.3 has been introduced in [119] as a mean-field analogue of sparsity and it is called *sparsity space constraint*. In the infinite-dimensional framework, the assumption that agents are indistinguishable is crucial for defining mean-field limits; therefore, the notion of sparsity as given in Section 2.2 loses its sense. The infinite-dimensional analogue of sparsity is the so-called *sparsity population constraint*

$$\int_{\omega(t)} d\mu(t) \leq c. \quad (2.16)$$

This constraint represents the idea of acting on a small part of the mass itself, and not on a small part of the configuration space, as we require in the space constraint in Definition 2.3. Even though the sparse population constraint is interesting from the theoretical point of view, it has a major drawback on the modeling point of view: when a measure is extremely concentrated, the constraint (2.16) implies that the control cannot act on the whole mass anymore. This is somehow unnatural, since a crowd that is already concentrated is the best configuration to steer. The sparsity space constraint given in Definition 2.3 is, on the other hand, the natural

definition of sparsity for mean-field equation of the form (2.15). The constraint is in some sense geographical, since the control can act only on a region of the space. The finite-dimensional analogue of such a control is the so-called *decentralized control*. A decentralized control acts based on partial information on the agents inside a certain neighborhood of the controlled ones. The decentralized control for multiagent systems is a well-established topic, see, for instance, [81, 140].

Given $D \subseteq \mathbb{R}^d$ we denote by $\mathcal{P}_c(D)$ the space of probability measures on $\mathbb{R}^d$ with compact support in D and by $\mathcal{P}_c^{ac}(D)$ the subspace of the subspace of $\mathcal{P}_c(D)$ of absolutely continuous measures (with respect to the Lebesgue measure). Given $R > 0$, B_R denotes the ball in $\mathbb{R}^d$ centered in 0 of radius R .

Definition 2.5. Given two probability measures μ and ν on $\mathbb{R}^d$ and $p \in [1, +\infty)$, the p -Wasserstein distance between μ and ν is

$$W_p(\mu, \nu) = \inf \left\{ \int_{\mathbb{R}^{2d}} |x - y|^p d\pi(x, y) \mid \pi \in \Pi(\mu, \nu) \right\}^{1/p},$$

where $\Pi(\mu, \nu)$ is the set of transference plans from μ to ν , i.e., of the probability measures π on $\mathbb{R}^d \times \mathbb{R}^d$ such that $\text{Proj}_x \# \pi = \mu$ and $\text{Proj}_y \# \pi = \nu$ with $\text{Proj}_x : (x, y) \mapsto x$ and $\text{Proj}_y : (x, y) \mapsto y$.

Assumption 2.1. The vector fields $f, g : \mathcal{P}_c(\mathbb{R}^d) \rightarrow \text{Lip}(\mathbb{R}^d, \mathbb{R}^d)$ satisfy

(i) there exists $R > 0$ such that if $\mu \in \mathcal{P}_c(B_R)$

$$f[\mu](x) = g[\mu](x) = 0, \quad \text{for every } x \notin B_R,$$

(ii) there exist $L > 0$, $Q > 0$ and $p \geq 1$ such that, for all $\mu, \nu \in \mathcal{P}_c(B_R)$ and for all $x, y \in \mathbb{R}^d$,

$$\begin{aligned} |f[\mu](x) - f[\mu](y)| &\leq L|x - y|, & |g[\mu](x) - g[\mu](y)| &\leq L|x - y|, \\ |f[\mu](x) - f[\nu](x)| &\leq QW_p(\mu, \nu), & |g[\mu](x) - g[\nu](x)| &\leq QW_p(\mu, \nu). \end{aligned}$$

Assumption 2.1 ensures existence and uniqueness of the solution for the Cauchy problem (2.15), for any admissible control u (see [29, 118] and Theorem 2.1 in [18]). In particular, thanks to Assumption 2.1(ii), the vector fields f and g generate semigroups. We denote by $e^{tf}\mu_0$ the unique solution of (2.14) at time t with initial data μ_0 and by $e^{tug}\mu_0$, $u \in \mathcal{U}$ the unique solution of the Cauchy problem associated with

$$\partial_t \mu + \nabla \cdot (ug[\mu]\mu) = 0, \tag{2.17}$$

with initial data μ_0 .

Given a Lyapunov function $V : \mathcal{P}_c(\mathbb{R}^d) \rightarrow \mathbb{R}$, the vector fields f and ug defined on the space $\mathbb{R}^d$ play the role of derivatives for the function $V[\mu]$, in the following sense: the vector field f induces an infinitesimal change in V that can be estimated as the derivative $\lim_{t \rightarrow 0} \frac{V[e^{tf}\mu] - V[\mu]}{t}$. For this limit to be well-defined we need to impose differentiability of the function $t \mapsto V[e^{tf}\mu]$ for any $\mu \in \mathcal{P}_c(\mathbb{R}^d)$. We can extend the notion of Lie derivative of a function along a vector field to the infinite-dimensional setting.

Definition 2.6. Assume that $V : \mathcal{P}_c(\mathbb{R}^d) \rightarrow \mathbb{R}$ is such that $t \mapsto V[e^{tf}\mu]$ is of class C^1 for all $\mu \in \mathcal{P}_c(\mathbb{R}^d)$. We define the *Lie derivative* of V along the vector field f as the limit

$$\mathcal{L}_f V[\mu] = \lim_{t \rightarrow 0} \frac{V[e^{tf}\mu] - V[\mu]}{t}. \quad (2.18)$$

Assumption 2.2. The vector fields $f, g : \mathcal{P}_c(\mathbb{R}^d) \rightarrow \text{Lip}(\mathbb{R}^d, \mathbb{R}^d)$ satisfy Assumption 2.1 and with the Lyapunov function $V : \mathcal{P}_c(\mathbb{R}^d) \rightarrow \mathbb{R}$ satisfy also the following

- (iii) $t \mapsto V[e^{tf}\mu]$ and $t \mapsto V[e^{tug}\mu]$ are C^2 for all $\mu \in \mathcal{P}_c(B_R)$ and $u \in \mathcal{U}$,
- (iv) the uncontrolled system is dissipative, i.e.,

$$\mathcal{L}_f V[\mu] \leq 0, \quad \text{for all } \mu \in \mathcal{P}_c(B_R),$$

- (v) for all μ^i and $\mu \in \mathcal{P}_c(\mathbb{R}^d)$ such that $\lim_{i \rightarrow +\infty} \int \phi d\mu^i = \int \phi d\mu$, for every $\phi \in C_c^0(\mathbb{R}^d, \mathbb{R})$, (weak convergence of measures) one has

$$\begin{aligned} \lim_{i \rightarrow +\infty} \mathcal{L}_f V[\mu^i] &= \mathcal{L}_f V[\mu], & \lim_{i \rightarrow +\infty} \mathcal{L}_f^2 V[\mu^i] &= \mathcal{L}_f^2 V[\mu], & \lim_{i \rightarrow +\infty} \mathcal{L}_{ug} V[\mu^i] &= \mathcal{L}_{ug} V[\mu], \\ \lim_{i \rightarrow +\infty} \mathcal{L}_{ug}^2 V[\mu^i] &= \mathcal{L}_{ug}^2 V[\mu], & \lim_{i \rightarrow +\infty} \mathcal{L}_f \mathcal{L}_{ug} V[\mu^i] &= \mathcal{L}_f \mathcal{L}_{ug} V[\mu], \end{aligned}$$

for every $u \in \mathcal{U}$.

Theorem 2.3 (Theorem 1.1 in [18]). *Let $f, g : \mathcal{P}_c(\mathbb{R}^d) \rightarrow \text{Lip}(\mathbb{R}^d, \mathbb{R}^d)$ and $V : \mathcal{P}_c(\mathbb{R}^d) \rightarrow \mathbb{R}$ satisfy Assumptions 2.2. Let*

$$\mathcal{Z} = \{\mu \in \mathcal{P}_c(\mathbb{R}^d) \mid \mathcal{L}_f V[\mu] = 0 = \mathcal{L}_{ug} V[\mu], \quad \text{for every } u \in \mathcal{U}\}.$$

Consider (2.15) with initial condition $\mu_0 \in \mathcal{P}_c^{ac}(B_R)$. Then there exists an admissible control such that the associated unique solution $\mu(t)$ of (2.15) satisfies $\mu(t) \in \mathcal{P}_c^{ac}(\mathbb{R}^d)$ for any $t \geq 0$, and $\mu(t)$ converges to $\mathcal{Z} \cap \mathcal{P}_c(B_R)$, i.e.

- $\lim_{t \rightarrow +\infty} \inf_{\nu \in \mathcal{Z} \cap \mathcal{P}_c(B_R)} W_p(\mu(t), \nu) = 0$,
or, equivalently,
- *there exists a choice $\nu(t) \in \mathcal{Z} \cap \mathcal{P}_c(B_R)$ for each $t \geq 0$ such that $\mu(t) \rightharpoonup \nu(t)$, i.e. $\lim_{t \rightarrow \infty} \int \phi d(\mu(t) - \nu(t)) = 0$, for all $\phi \in C_c^\infty(\mathbb{R}^d)$.*

Theorem 2.3 provides an explicit control strategy to steer a kinetic multiagent model to consensus, represented by the invariant set $\mathcal{Z}$. The strategy have been used in [119] for the control of the kinetic Cucker–Smale model (2.6) in the simple case with constant g and for the control of the kinetic Hegselmann–Krause model (2.2)-(2.3) in Section 6 of [18].

Remark 2.7. The main idea of the proof of Theorem 2.3 is to design a control which makes V decrease along trajectories. The control is chosen with a steepest descent method, in analogy with the finite-dimensional approach described in Section 2.2.1. The resulting control steers the system to the set $\mathcal{Z}$ on which the Lie derivatives of V vanish. The construction of the control is explicit and it is based on reducing the set of admissible control, which is infinite-dimensional, to a finite-dimensional set by looking for controllers depending on three parameters $a = (a_1, \dots, a_d), b = (b_1, \dots, b_d) \in \mathbb{R}^d$, and $\eta > 0$. Consider a Lipschitz function $v = v(a, b, \eta)$ with support in $\omega(a, b, \eta) := [a_1 - \eta, b_1 + \eta] \times [a_2 - \eta, b_2 + \eta] \times \dots \times [a_d - \eta, b_d + \eta]$ such that $v(a, b, \eta)(x) = 1$ for $x \in [a_1, b_1] \times [a_2, b_2] \times \dots \times [a_d, b_d]$. Define the *slope function*

$$s_t(a, b, \eta) = |\mathcal{L}_{v(a, b, \eta)g[\mu(t)]}V[\mu(t)]|,$$

describing the instantaneous variation of V in $\mu(t)$ under the action of the function $v(a, b, \eta)$. Apply a steepest descent method by choosing the control corresponding to one of the maximizers (a^*, b^*, η^*) of s_t in the set

$$\Omega_t = \{(a, b, \eta) \mid |\omega(a, b, \eta)| \leq c \text{ and } \eta \geq t^{-1}\}. \quad (2.19)$$

Finally, the control is of the form

$$u(t, x) = -\chi_{\omega(a^*, b^*, \eta^*)}v(a^*, b^*, \eta^*)\text{sign}(\mathcal{L}_{v(a^*, b^*, \eta^*)g[\mu(t)]}V[\mu(t)]).$$

The condition $|\omega(a, b, \eta)| \leq c$ in (2.19) ensures that the space constraint of Definition 2.3 is satisfied. The condition $\eta \geq t^{-1}$ implies that the control function is uniformly Lipschitz for any bounded interval of time, thus ensuring that $\mu(t)$ remains absolutely continuous with respect to the Lebesgue measure. Choosing the control as the instantaneous maximizer of s_t may cause chattering phenomena, as it has been already noticed in finite dimension in Section 2.2.1. For this reason, the control is regularized by means of an *hysteresis* in analogy with Strategy 3.

2.3 Examples and extensions

2.3.1 Global stabilization and optimal control of the Cucker–Smale model

In [15, 14] we consider the controlled version of the Cucker–Smale system (2.4), that is

$$\begin{cases} \dot{x}_i(t) &= v_i(t), \\ \dot{v}_i(t) &= \frac{1}{N} \sum_{j=1}^N a(\|x_j(t) - x_i(t)\|)(v_j(t) - v_i(t)) + u_i(t), \end{cases} \quad (2.20)$$

for $i = 1, \dots, N$, and $x_i \in \mathbb{R}^d$, $v_i \in \mathbb{R}^d$. Admissible controls are $u = (u_1, \dots, u_N) \in (\mathbb{R}^d)^N$ such that

$$\sum_{i=1}^N \|u_i(t)\| \leq M, \quad (2.21)$$

for some given positive constant M .

System (2.20) fits within the framework of Theorem 2.2 which provides global stabilization to a neighborhood of consensus. More precisely Strategy 2 gives a sparse feedback u° defined as follows. Let $v_{\perp_i} = v_i - \frac{1}{N} \sum_{k=1}^N v_k$ for $i = 1, \dots, N$ and

$$\gamma(X) = \int_{\sqrt{X}}^{+\infty} a(\sqrt{2Nr}) dr.$$

- If $\max_{1 \leq i \leq N} \|v_{\perp_i}(t)\| \leq \gamma(X(t))^2$, then $u^\circ(t) = 0$.
- If $\|v_{\perp_j}(t)\| = \max_{1 \leq i \leq N} \|v_{\perp_i}(t)\| > \gamma(X(t))^2$ (with j be the smallest index) then

$$u_j^\circ(t) = -M \frac{v_{\perp_j}(t)}{\|v_{\perp_j}(t)\|}, \quad \text{and} \quad u_i^\circ(t) = 0 \quad \text{for every } i \neq j.$$

The threshold γ is given by Proposition 2.1 and guarantees that the control switches off when the solution enters the flocking region. Theorem 2.2 then reads as follows.

Theorem 2.4 (Theorem 2.3 in [14]). *For every initial condition $(x_0, v_0) \in (\mathbb{R}^d)^N \times (\mathbb{R}^d)^N$ there exists $\tau_0 > 0$ such that for all $\tau \in (0, \tau_0]$ the sampling solution of (2.20) associated with the control u° , the sampling time τ , and the initial pair (x_0, v_0) reaches the consensus region in finite time.*

Remark 2.8 (Sparse is “optimal”). It has been proven in Proposition 3.1 in [14] that, “sparse is better” in the sense that:

For every time t , $u^\circ(t)$ minimizes $\frac{d}{dt}V(t)$ over all possible feedback controls.

In other words, at every instant of time t , the above feedback control $u^\circ(t)$ is the best choice in terms of the rate of convergence to flocking. This means that a policy maker who is not allowed to have prediction on future developments should always consider more favorable to intervene with stronger actions on the fewest possible instantaneous optimal leaders, rather than trying to control more agents with minor strength.

It is possible to prove that, generically, any first-order or second-order model is locally controllable near the consensus/alignment manifold by means of a sparse control. More precisely, given a generic pair of configurations sufficiently close to consensus, there exists a control with at most one non-zero component steering the system from one configuration to the other. This property was proven in Proposition 4.1 in [14] for alignment systems. The proof relies on the fact that given a generic Laplacian matrix L satisfying some open condition on the coefficients, and any column vector B with only one component different from 0, the linear system

$$\dot{x} = -Lx + Bu$$

verifies the Kalman rank condition for controllability. Moreover, it is possible to choose the controlled agent *a priori*. By stabilization and iterated local controllability along a path of consensus points (note that the set of consensus points is arc-connected), we obtain the following global sparse controllability result.

Corollary 2.5 (Corollary 4.1 in [14]). *Any point of $(\mathbb{R}^d)^N \times (\mathbb{R}^d)^N$ can be steered to almost any point of the consensus manifold in finite time by means of a sparse control.*

Remark 2.9 (Sparse optimal control). Consider the problem of minimizing the cost functional

$$\int_0^T \sum_{i=1}^N \left(v_i(t) - \frac{1}{N} \sum_{j=1}^N v_j(t) \right)^2 dt + \gamma \sum_{i=1}^N \int_0^T \|u_i(t)\| dt$$

with free final point, where $\gamma > 0$ is fixed, under the dynamical constraint (2.20), and under the control constraint (2.21). The ℓ^1 -norm term in the cost (with weight γ) promotes sparsity features of the optimal control. The problem is studied in [15, 14] using the Pontryagin maximum principle and by developing genericity arguments. However, because of the coupling between space and velocity and the high dimensionality of the problem, these sparsity features are not easy to check in practice in this case.

2.3.2 Optimal control via leader of social dynamics

The Hegselmann–Krause model

In [23] we consider the control of the HK opinion formation model (2.1) with leadership. The system of N interacting agents with one leader with opinions in $\mathbb{R}^d$ is given by

$$\begin{cases} \dot{x}_0(t) = u(t), \\ \dot{x}_i(t) = \sum_{j=1}^N a_{ij}(x_j(t) - x_i(t)) + c_i(x_0(t) - x_i(t)), \quad \text{for } i = 1, \dots, N, \end{cases} \quad (2.22)$$

for given initial positions $x_j(0) \in \mathbb{R}^d$, $j = 0, 1, \dots, N$. The state of the system, representing the agents' and the leader's opinions is $x = (x_0, x_1, \dots, x_N) \in \mathbb{R}^{(N+1)d}$. We denote the leader by the index 0 and the N agents by the index $i = 1, \dots, N$. The control satisfies the constraint $\|u\| \leq M$, for some given $M > 0$ and acts only on the leader. We set $a_{ij} = a(\|x_i(t) - x_j(t)\|)$, for a smooth cut-off function

$$a(r) = \begin{cases} 1, & 0 \leq r \leq \delta, \\ \varphi(r), & \delta < r < (\delta + \varepsilon), \\ 0, & (\delta + \varepsilon) \leq r, \end{cases} \quad (2.23)$$

where δ is the bounded confidence, $\varphi(r)$ is a decreasing smooth function on $(\delta, \delta + \varepsilon]$, and $\varepsilon > 0$ is a parameter of the HK model defining the width of the region where the cut-off function decays to zero. The second term in the dynamics (2.22) models the action of the leader on agent i . A leader is an agent not influenced by the other agents' opinions. The parameter $c_i = \phi(\|x_i - x_0\|)$, represents the rate of relationship between the leader and the other agents, where $\phi : [0, \infty) \rightarrow [0, 1]$ is a smooth non-increasing non-negative function such that $\phi(0) = 1$ and $\lim_{r \rightarrow \infty} \phi(r) = 0$. In other words the leader has the ability to influence every agent with a factor that is inversely proportional to its distance from the agent.

Global stabilization is ensured by Theorem 2.2. More precisely we have the following result.

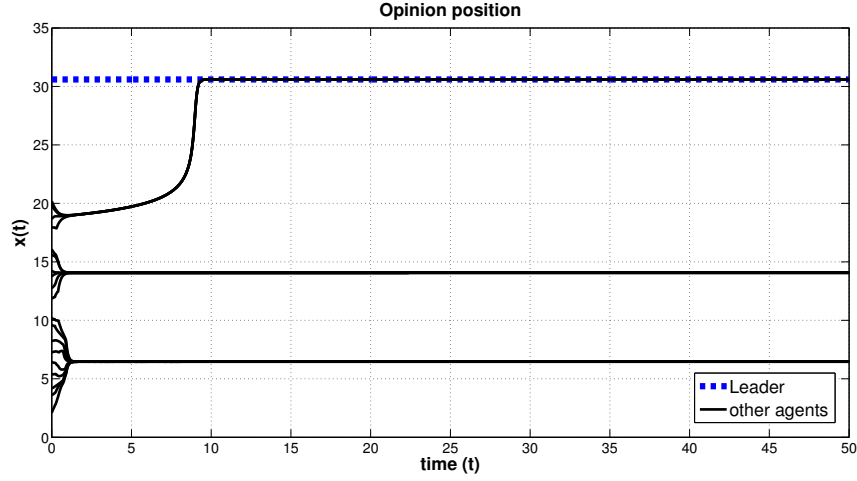


Figure 2.2: Free evolution of the system (uncontrolled case) with $N + 1 = 20$ agents, initial positions randomly chosen in $[-20, 20]$, and leader's initial position $x_0(0) = 30$. The evolution of the agents' opinions is represented by solid lines and the dashed line represent the leader's opinion.

Theorem 2.6 (Theorem 1 in [23]). *For every initial condition $x(0) \in \mathbb{R}^{(N+1)d}$ and every M there exists a feedback $x \mapsto u(x) \in \mathbb{R}^d$ with $\|u\| \leq M$ such that the associated solution tends to consensus.*

Notice that in this case the feedback can be chosen to be continuous.

In contrast with system (2.20), system (2.22) is not locally controllable. Indeed, in this case, system (2.22) is linear whenever the agents are sufficiently close to consensus and it does not verify the rank condition (see [23], Section 4), necessary for controllability. For more general situations the local controllability depends on the communication rate ϕ between the leader and the agents and, a priori, we cannot infer the local controllability of the system.

However the system verifies a “weaker” local controllability property, that is the local controllability to consensus. Indeed if the agents are sufficiently close to each other they are attracted by the leader. This fact is a consequence of the local attractivity of the leader's state x_0 . As a consequence we have the following result of global stabilization and partial controllability.

Corollary 2.7 (Corollary 1 in [23]). *For every initial condition $x(0)$ and for every consensus x^* there exists a control u such that the associated solution $x(t)$ of (2.22) with initial condition $x(0)$ tends to x^* .*

Once global stabilization is achieved it is natural to study how to enforce consensus in the “best” possible way. We consider the optimal control problems of minimizing the cost functional

$$J(x, u) := \frac{1}{2N^2} \int_0^T \sum_{i,j=1}^N |x_i(t) - x_j(t)|^2 dt + \frac{1}{2} \int_0^T \sum_{i=1}^N |x_0(t) - x_i(t)|^2 dt + \frac{1}{2} \int_0^T |u(t)|^2 dt, \quad (2.24)$$

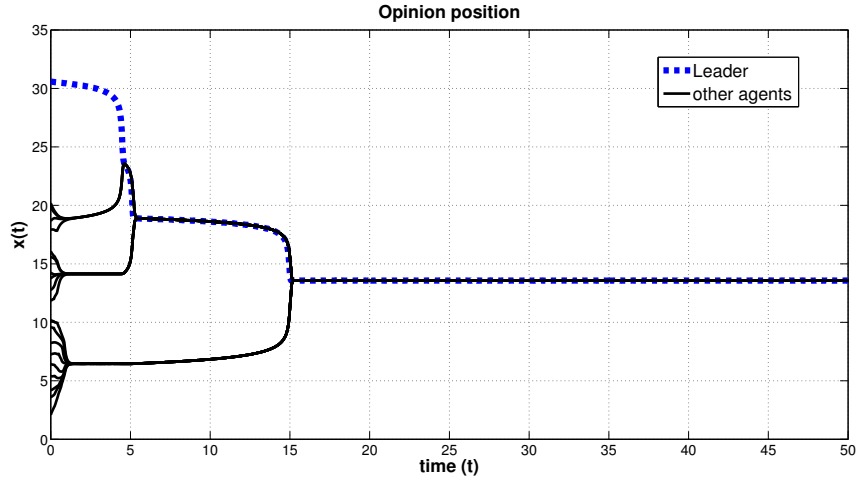


Figure 2.3: Stabilizing control with $N + 1 = 20$ agents, initial positions randomly chosen in $[-20, 20]$, and leader's initial position $x_0(0) = 30$. The evolution of the agents' opinions is represented by solid lines and the dashed line represent the leader's opinion.

under dynamical constraint (2.22).

In [23] we propose a Runge–Kutta (RK) discretization scheme for solving the corresponding optimal control problem (2.22)–(2.24). We use a direct discretization method and we consider a model predictive control (MPC) scheme to construct a feedback control strategy. In our experiments, the connectivity functions $a_{ij} = a(\|x_i - x_j\|)$ are given by (2.23) with $\varphi(r) = \left(1 + \tanh\left(\frac{1}{r-\delta} + \frac{1}{r-(\delta+\varepsilon)}\right)\right)/2$, $\delta = 2.5$, $\varepsilon = 0.05$, and $\phi(\|x_i - x_0\|) = 10e^{-\frac{\|x_i - x_0\|^2}{20}}$.

Figure 2.2 shows the numerical results. The initial positions of the 19 agents' opinion are distributed randomly in $[-20, 20]$ while the leader's initial opinion is $x_0(0) = 30$. The opinions $x_i(t)$ for $i = 0, \dots, N$, converge to a limiting opinion x^* and, due to the limitation of confidence, clusters of agents' opinions are formed. The action of the optimal control is depicted in Figure 2.4. The initial opinions of the agents are randomly chosen in $[-5, 10]$ and the opinion of the leader is at $x_0(t_0) = 20$. Figure 2.3 displays the action of the stabilizing control given by Theorem 2.6.

The Heider Balance model

In [22] we studied control via leader of a model of evolving social network, called the Heider Balance model. The balance theory proposed by F. Heider [86, 87] models how people develop their relationship with other people and with things in their environment based on a cognitive consistency motive that drives toward psychological balance. This motive urges to maintain one's values and beliefs over time resulting in the preference to have a balanced state where the affect valence in the system multiplies out to a positive result. Specifically, in the relation of three people, balance state occurs when the product of all signs of sentiment relations is positive. That is, balance state occurs when there are sentiment relations with signs all positive or two

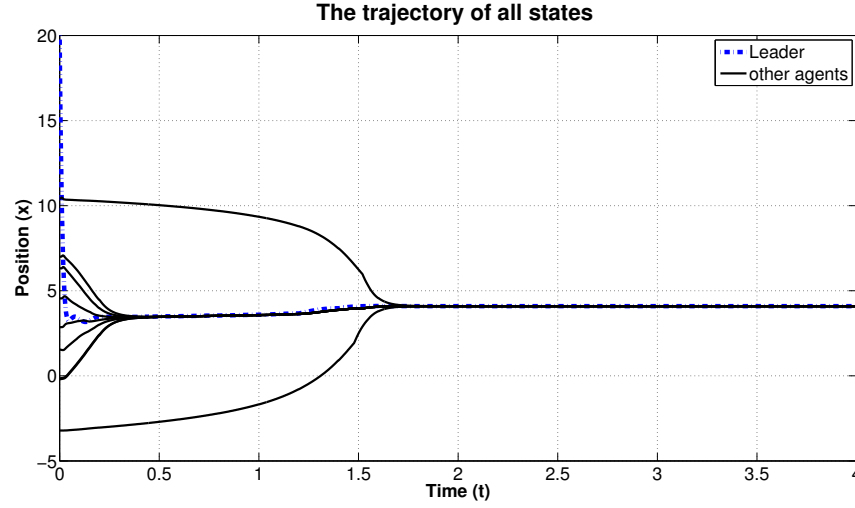


Figure 2.4: Optimal control. Simulation with $N + 1 = 10$ agents. The goal is to force all agents to consensus. The evolution of the agents' opinions is represented by solid lines and the dashed line represent the leader's opinion.

negatives and one positive. We refer to sentiment relation between two people as a linking edge to which a positive value is associated in the case of friendship and a negative value in the case of hostility.

We consider an optimal control problem based on the continuous-time Heider balance model proposed by [100]. In this problem, an additional reference agent, or *leader*, enters in the network with partial or full connection to all other agents of the network. This agent acts on the network by modifying the values of the edges that connect to it. The continuous-time Heider balance (HB) model is given by

$$\dot{x}_{ij} = c(x_{ij}(t); R) \sum_{\substack{k=1 \\ k \neq i, j}}^N x_{ik} x_{kj}, \quad \text{for } i, j = 1, \dots, N, \text{ and } i \neq j, \quad (2.25)$$

with given initial conditions $x_{ij}(t_0) = x_{ij}^0$ and

$$c(x_{ij}; R) = \frac{1}{N-2} \left(1 - \frac{x_{ij}^2}{R^2} \right), \quad \text{for } R > 0.$$

The indices i, j represent individuals in the network and $x_{ij} \in \mathbb{R}$ denotes the relationship between agent i and j . A positive value of x_{ij} determines friendship between agent i and j ; conversely, a negative value of x_{ij} expresses hostility. The relationship is symmetric, that is $x_{ij} = x_{ji}$ for any i and j . Notice that in the dynamics given by (2.25), $x_{ii} = 0$ and each agent is connected to all agents in the network, that is, social structure can be seen as fully connected graph.

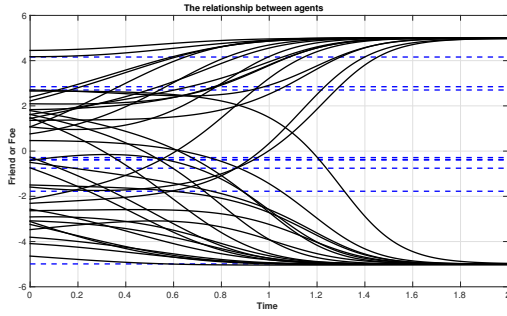
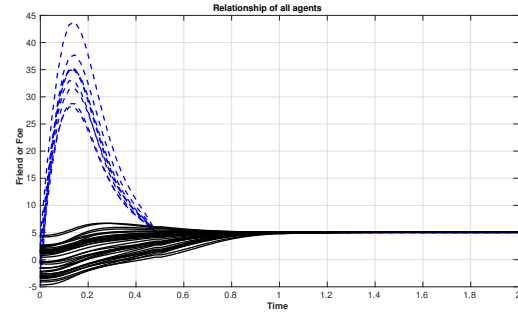
(a) $x_{ij}(0) \in [-5, 5]$. Free evolution(b) $x_{ij}(0) \in [-5, 5]$. Controlled evolution

Figure 2.5: Simulation with $N + 1 = 10$ agents. In Figure(a) the initial relations of friendship and hostility are randomly distributed in $[-5, 5]$. Figure(a) shows the result of the free evolution of the system while Figure(b) shows the action of the controllers. The dashed lines represent the relationship of the leader with the agents. Relationships between agents are solid lines.

As showed in [22], if the relationship value of an edge starts with a value greater than or equal to $-R$, then the HB model reaches a balanced state where the trajectories of relationships split into two groups, one of them asymptotically reaches the value R and the other the opposite value $-R$.

The optimal control problem for the HB model with the presence of a leader is to minimize

$$J(x, u) = \frac{1}{2} \sum_{i=1}^N (x_{ij}(T) - R)^2 + \frac{\nu}{2} \int_0^T \|u(t)\|^2 dt \quad (2.26)$$

subject to the differential constraints given by

$$\begin{cases} \dot{x}_{0i}(t) = u_i(t) \\ \dot{x}_{ij}(t) = c(x_{ij}; R) \sum_{\substack{k=1 \\ k \neq i, j}}^N x_{ik}x_{kj} + \gamma x_{i0}x_{j0}, & \text{for } i, j = 1, \dots, N, \text{ and } i \neq j, \end{cases}$$

with initial conditions $x_{ij}(0) = x_{ij}^0$, for $i, j = 1, \dots, N$. We solve numerically the optimal control problem by adapting the RK discretization scheme proposed in [23] and [53]. For the simulation we fix a time horizon $T = 2$. The goal is to find the optimal control driving the HB system to a friendship state where $x_{ij} = R$ for all $i, j = 0, \dots, N$, $i \neq j$. We consider two series of experiments; in the first one (see Figure 2.5), the initial conditions $x_{ij}(0) \in [-R, R]$. In the second one (see Figure 2.6), the state of relations starts with hostility that is characterized by values in a neighborhood of the unstable equilibrium point $-R$. In both cases, we solve the optimal control problem (2.26) with $N = 9$ agents and one leader and with $R = 5$. In the objective functional we take $\nu = 0.001$. To apply the MPC strategy, the time horizon is divided into subintervals of size $\Delta t = 0.25$.

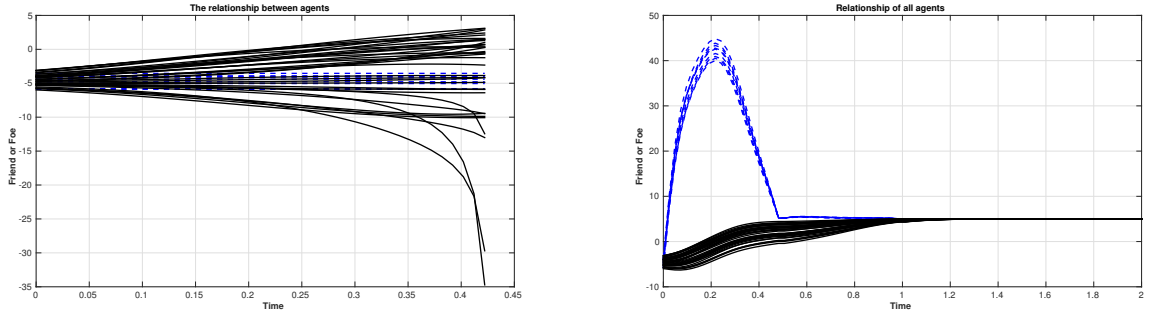
(a) $x_{ij}(0) \in [-6, -3]$. Free evolution(b) $x_{ij}(0) \in [-6, -3]$. Controlled evolution

Figure 2.6: Simulation with $N + 1 = 10$ agents. In Figure(a) the initial relations of friendship and hostility are randomly distributed in $[-6, -3]$. Figure(a) shows the result of the free evolution of the system while Figure(b) shows the action of the controllers. The dashed lines represent the relationship of the leader with the agents. Relationships between agents are solid lines.

2.4 Avoiding chattering in optimal control

In control theory, the term chattering is used to denote fast oscillations of controls, such as an infinite number of switchings over a finite time interval. The problem naturally arise when dealing with the sparse control of multiagents systems as pointed out in Section 2.2.1. However chattering phenomena are a common issue in optimal control. The first example is the well-known Fuller's problem [79]. It was later shown in [101] that chattering behavior is rather typical, indeed the set of single-input optimal control problems which have a control-affine Hamiltonian and whose solution is chattering is an open semi-algebraic set (see also [142]).

In [16] we consider a general regularization procedure for non-linear optimal control problems of the form

$$\begin{cases} \min_{u \in \mathcal{U}} \int_0^{t(u)} L(s, x(s), u(s)) ds, \\ \dot{x} = F(x, u), \quad u \in \mathcal{U}, \\ x(0) = x_0, \quad x(t(u)) = 0, \end{cases} \quad (2.27)$$

where $F \in C^\infty(\mathbb{R}^N \times \mathbb{R}^m, \mathbb{R}^N)$, $F(0, 0) = 0$,

$$\mathcal{U} = \{u(\cdot) \text{ measurable} \mid u(t) \in U \text{ for a.e. } t\}, \quad (2.28)$$

with $U \subset \mathbb{R}^m$ measurable and containing 0, with $x_0 \in \mathbb{R}^N$, and with Lagrangian $L \in C^0(\mathbb{R} \times \mathbb{R}^N \times \mathbb{R}^m)$. The final time in (2.27) may be fixed or free. If it is a fixed $T > 0$, then one has to replace $t(u)$ by T in what follows.

Definition 2.10. By *chattering* control we mean a measurable function $u : [0, t(u)] \rightarrow U$ such that there exists an increasing sequence $\{t_n\}_{n \in \mathbb{N}}$ converging to $t(u)$ with the property that

$\text{TV}(u|_{[0,t_n]}) < +\infty$ for every $n \in \mathbb{N}$, and

$$\lim_{n \rightarrow +\infty} \text{TV}(u|_{[0,t_n]}) = +\infty.$$

Assumption 2.3. (i) For every $(t, x) \in \mathbb{R} \times \mathbb{R}^N$, the set

$$\{(F(x, u), L(t, x, u) + \gamma) \mid u \in U, \gamma \geq 0\}$$

is convex;

(ii) the control set U is compact, and there exists $b > 0$ such that $t(u) + \|x_u(\cdot)\|_\infty \leq b$, for every $u \in \mathcal{U}$ steering x_0 to 0 in time $t(u)$,

Assumption 2.3 guarantees the existence of at least one optimal solution $x^*(\cdot)$, associated with a control $u^* : [0, t(u^*)] \rightarrow \mathbf{U}$, for the optimal control problem (2.27). The first assumption, Assumption 2.3.(i) means that the epigraph of extended velocities is convex and it is satisfied, for example, by control-affine systems with control-affine or quadratic cost.

The idea is to penalize chattering controls by adding a total variation term to the cost functional in (2.27) with a small weight ε . Given any $\varepsilon \geq 0$, we consider the optimal control problem

$$\begin{cases} \min_{u \in \mathcal{U}} \left(\int_0^{t(u)} L(s, x(s), u(s)) ds + \varepsilon \text{TV}(u) \right), \\ \dot{x} = F(x, u), \quad u \in \mathcal{U}, \\ x(0) = x_0, \quad x(t(u)) = 0. \end{cases} \quad (2.29)$$

Denote by $\mathcal{F} = \{F(\cdot, u) : \mathbb{R}^N \rightarrow \mathbb{R}^N \mid u \in \mathbf{U}\}$, the family of vector fields associated with the dynamics of the control system

$$\dot{x} = F(x, u), \quad u \in \mathcal{U}, \quad (2.30)$$

Theorem 2.8 (Theorem 1 in [16]). *Let system (2.27) satisfy Assumption 2.3. Assume that $\text{Lie}_0 \mathcal{F} = \mathbb{R}^N$, and the control system (2.30) is small-time locally controllable at 0. Then, for every $\varepsilon > 0$, the optimal control problem (2.29) has at least one solution. Moreover, for every optimal solution $x_\varepsilon(\cdot)$ of (2.29), associated with a control $u_\varepsilon : [0, t(u_\varepsilon)] \rightarrow U$, we have*

$$\lim_{\varepsilon \rightarrow 0} \int_0^{t(u_\varepsilon)} L(t, x_\varepsilon(t), u_\varepsilon(t)) dt = \int_0^{t(u^*)} L(t, x^*(t), u^*(t)) dt. \quad (2.31)$$

Theorem 2.8 establishes the existence of a non-chattering control u_ε which is quasi-optimal for (2.27) in the sense that the cost of u_ε converges to the optimal value of (2.27).

Remark 2.11. In general one cannot infer convergence of the optimal solution $x_\varepsilon(\cdot)$ of (2.29) to an optimal solution of the original problem (2.27). However, if the optimal solution x^* of (2.27) is unique then one has that an optimal solution $x_\varepsilon(\cdot)$ of (2.29) converges uniformly to $x^*(\cdot)$.

Remark 2.12. The Lie algebra and small-time controllability assumptions, although generic, may be slightly weakened without altering the conclusion of the theorem: they can be replaced by assuming local controllability in a neighborhood of the origin in arbitrarily small time and with piecewise constant controls.

Theorem 2.8 establishes the convergence of costs (2.31). It is then interesting to derive a speed of convergence. To this aim we need the following “strong” notion of controllability which requires a uniform bound on the total variation of the control and a steering time comparable with the minimum time.

Assumption 2.4. The control system (2.30) is small-time locally controllable at 0 and there exists a neighborhood $\mathcal{N}$ of 0 and $M \geq 1$ such that every $y \in \mathcal{N}$ can be steered to 0 in time $\tau_y \leq M\mathcal{T}(y)$ by means of a control u satisfying $\text{TV}(u) \leq M$.

Here $x_0 \mapsto \mathcal{T}(x_0)$ is the time-optimal map associated with the control system (2.30), defined by

$$\mathcal{T}(x_0) = \inf\{t > 0 \text{ such that } \dot{x} = F(x, u), x(0) = x_0, x(t) = 0, u \in \mathcal{U}\}. \quad (2.32)$$

Theorem 2.9 (Theorem 2 in [16]). *Assume that system (2.27) satisfies Assumption 2.3 and that the control system (2.30) satisfies Assumption 2.4. Assume, moreover, that:*

- (i) *the optimal control problem (2.27) admits an optimal solution u^* which is chattering and whose sequence of switching times $(t_n)_{n \in \mathbb{N}}$ satisfies $(t(u^*) - t_n) = O(n^{-\beta})$ for some $\beta > 0$;*
- (ii) *the time-optimal map is $C^{0,\alpha}$ for some $\alpha \in (0, 1]$ in a neighborhood of 0.*

Then, for every $\varepsilon > 0$, the optimal control problem (2.29) has at least one solution. Moreover, for every optimal solution $x_\varepsilon(\cdot)$ of (2.29), associated with a control $u_\varepsilon : [0, t(u_\varepsilon)] \rightarrow \mathbf{U}$, we have

$$\int_0^{t(u_\varepsilon)} L(t, x_\varepsilon(t), u_\varepsilon(t)) dt - \int_0^{t(u^*)} L(t, x^*(t), u^*(t)) dt = O\left(\varepsilon^{\frac{\alpha\beta}{1+\alpha\beta}}\right).$$

Remark 2.13. For linear control systems and for driftless control-affine systems, Assumption 2.4 is related to controllability. Sufficient conditions guaranteeing that Assumption 2.4 holds true can be found in [131, 138] for single-input control systems. For more general control-affine systems, Assumption 2.4 is related to the *Exact State Space Linearizability Problem* (see Appendix A in [16]).

Remark 2.14. Assumption (i) is verified for a large class of systems having an exponential rate of accumulation of switchings (see [101]). In this case the convergence rate is $O(\varepsilon^\gamma)$ for every $\gamma < 1$.

Remark 2.15. Sufficient conditions for assumption (ii) have been established in [47, Theorem 3.3, 3.10, 3.12], where the authors provide an estimate on the Hölder exponent $\alpha \in (0, 1]$.

Remark 2.16. Theorem 2.8 extends to optimal control problems involving state-constraints, under adequate controllability assumptions, Theorem 3 in [16] provides a regularization result for optimal trajectories having an infinite sequence of contact points with the constraint's boundary (Robbins phenomenon [124]). The penalization term is the total variation of the indicator function of the boundary, which, essentially, counts the contact points with the constraint's boundary. The result extends as well to hybrid optimal control problems. Indeed Theorem 4 in [16] provides a convergence result to regularize the Zeno phenomenon (see, for instance, [143]), obtaining estimates of the cost convergence as the number of location switchings grows.

2.5 Modeling

Beside controllability we used geometric techniques to address the modeling problem for multi-agent systems. In particular we focused on opinion formation and on pedestrian models.

2.5.1 Opinion formation on a manifold

In [17] we consider the problem of modeling opinion formation. Several mathematical models have been proposed in the last few years. Among the most widely used we cite the already mentioned model by Hegselmann and Krause [85], presented in Section 2.1.1, we cite the Sznajd model [139], the Déffuant model [72], and the classical Voter model.

All these models present very interesting features but also lack of connections with real life examples. This issue has been thoroughly addressed in [134]. Describing the complex interactions between individuals with simple mathematical rules is a very hard task, nevertheless the models above present features that are good starting points to develop more complete models. In [17] we present an analysis of strengths and weaknesses of the models in literature. Thanks to a wide literature it is possible to find interesting features with strong sociological motivations. We combine these features with a complex dynamics. To this aim we propose a non-linear model on the d -dimensional sphere, whose dynamics is based on the projection of the linear dynamics in the ambient space onto the tangent space of the manifold. Given N interacting agents whose opinions are represented by $x_1, \dots, x_N$ on the $\mathbb{S}^d$, the time evolution of the opinion x_i of agent i is governed by

$$\dot{x}_i = \sum_{j=1}^N a_{ij}(x_j - \langle x_i, x_j \rangle x_i), \quad x_i \in \mathbb{S}^d, i = 1, \dots, N. \quad (2.33)$$

The RHS is the projection of the vector x_j on the orthogonal to x_i . In [17] we consider constant and variable coefficients a_{ij} with no assumptions on the symmetries of the network, including symmetric interactions $a_{ij} = a_{ji}$, antisymmetric interactions $a_{ij} = -a_{ji}$ as well as general coefficients.

The non-linearity of the state space yield a rich structure for the dynamics leading to unusual equilibria. Indeed three equilibrium configuration are present: consensus, antipodal equilibria,

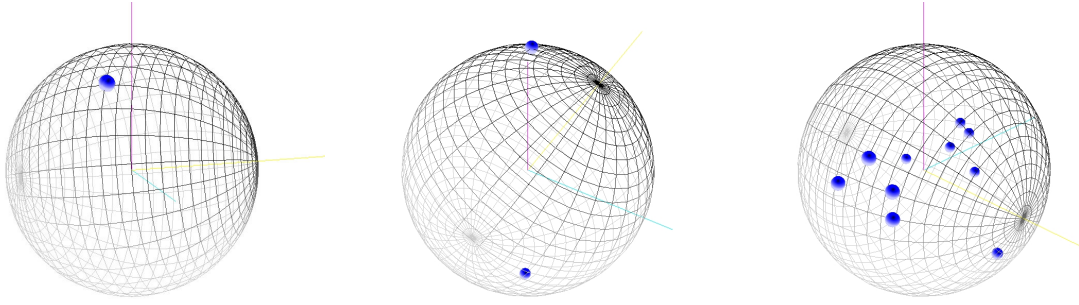


Figure 2.7: Examples of equilibria on $\mathbb{S}^2$. Larger points are closer to the viewer and smaller points are on the opposite side of the sphere.

and polygonal equilibria (see Figure 2.7), in contrast with linear system having consensus as sole possible equilibrium. This allows a better representation of the complex reality of opinion formation in which consensus is not the only steady configuration. Moreover another interesting feature of model (2.33) is the existence of “dancing equilibria”: steady configuration with constant distance between the opinions.

2.5.2 Psychological bias in pedestrian motion

Pedestrian models represent a very interesting research field. This domain is particularly challenging because of the complex interactions between the pedestrians and the environment. Current mathematical models partially accomplish this goal, one of the main limitations being the lack of psychological insight. Indeed, most of the mathematical models use a mechanistic approach assuming that pedestrians only move according to physical principles, such as the least action principle or the minimization of the path length and behavioral models have been proposed only in the last few years [43, 40]. In [13] we analyse pedestrian behaviors gathering results from two experiments on Rutgers-Camden campus, involving up to 8500 pedestrians. In the first experiment, the results show the strong influence that the knowledge of being in an experiment has on the behavior of pedestrians, far beyond random fluctuations. In the second one, we determine the cognitive and mechanical aspects involved in human locomotion. From the results of the experiments we propose a cognitive-mathematical model combining psychological perceptions and models of optimization in robotics (for example [30]).

2.6 Perspectives

The topic of complex systems represents a interesting research field for the quantity and the variety of models and related challenges. Control problems naturally arise at every scale: from systems of ordinary differential equations to partial differential equations.

A new problem at the microscopic scale is represented by the analysis of the influence of several leaders (represented by several controls), in cooperation or in competition, on the evolution of a finite number of interacting agents. The goal is to study the steady states of

such a system and their nature in order to describe Nash equilibria, Pareto equilibria, etc. This kind of analysis allows a better comprehension of emergent phenomena in opinion formation (as defined in Section 2.1.1), in the evolution of financial markets [25] or in crowd behavior [88]. The idea is to adapt methods of sparse stabilization (Section 2.2.1) and of control via leader (Section 2.3.2) to the context of differential games.

Models coupling microscopic and macroscopic scale represent a promising area of research. This kind of multiscale models are used mostly for crowd dynamics [68]. The extension of geometric techniques of sparse control to coupled ODE-PDE multiscale models provided partial results of controllability for systems obtained as mean-field limit of a particular alignment model [77]. However, the general control problem is far from being completely understood. The idea would be to couple the control technique used in the sparse stabilization of dissipative systems of Section 2.2.1 and in the control of mean-field transport equations of Section 2.2.2 to study controllability and asymptotic behavior of general multiscale models.

Bibliography

References by Marco Caponigro

- [1] A. Aydoğdu, M. Caponigro, S. McQuade, B. Piccoli, N. Pouradier Duteil, F. Rossi, and E. Trélat. “Interaction Network, State Space, and Control in Social Dynamics”. In: *Active Particles, Volume 1 : Advances in Theory, Models, and Applications*. Ed. by N. Bellomo, P. Degond, and E. Tadmor. Cham: Springer International Publishing, 2017, pp. 99–140.
- [2] U. Boscain, M. Caponigro, T. Chambrion, and M. Sigalotti. “A weak spectral condition for the controllability of the bilinear Schrödinger equation with application to the control of a rotating planar molecule”. In: *Communications in Mathematical Physics* 311.2 (2012), pp. 423–455.
- [3] U. Boscain, M. Caponigro, and M. Sigalotti. “Multi-input Schrödinger equation: Controllability, tracking, and application to the quantum angular momentum”. In: *Journal of Differential Equations* 256.11 (2014), pp. 3524–3551.
- [4] N. Boussaïd, M. Caponigro, and T. Chambrion. “Weakly-coupled systems in quantum control”. In: *IEEE Trans. Automat. Control* 58.9 (2013), pp. 2205–2216.
- [5] N. Boussaïd, M. Caponigro, and T. Chambrion. “An approximate controllability result with continuous spectrum : the Morse potential with dipolar interaction”. In: *SIAM Conference on Control and its applications*. Paris, France, July 2015.
- [6] N. Boussaïd, M. Caponigro, and T. Chambrion. “Approximate controllability of the Schrödinger equation with a polarizability term”. In: *CDC - The 51st IEEE Conference on Decision and Control - 2012*. Maui, Hawaii, United States, Dec. 2012, pp. 3024–3029.
- [7] N. Boussaïd, M. Caponigro, and T. Chambrion. “Efficient finite dimensional approximations for the bilinear Schrödinger equation with bounded variation controls”. In: *21st International Symposium on Mathematical Theory of Networks and Systems (MTNS2014)*. Groningen, Netherlands, July 2014, pp. 1889–1891.
- [8] N. Boussaïd, M. Caponigro, and T. Chambrion. “Energy Estimates for Low Regularity Bilinear Schrödinger Equations”. In: *IFAC Proceedings Volumes* 46.26 (2013). 1st IFAC Workshop on Control of Systems Governed by Partial Differential Equations, pp. 25–30.
- [9] N. Boussaïd, M. Caponigro, and T. Chambrion. “On the Ball–Marsden–Slemrod obstruction in bilinear control systems”. Preprint hal-01537743. June 2017.
- [10] N. Boussaïd, M. Caponigro, and T. Chambrion. “Regular propagators of bilinear quantum systems”. Preprint hal-01016299. Oct. 2017.
- [11] N. Boussaïd, M. Caponigro, and T. Chambrion. “Total Variation of the Control and Energy of Bilinear Quantum Systems”. In: *Conference on Decision and Control*. Florence, Italy, Dec. 2013, pp. 3714–3719.

- [12] N. Boussaïd, M. Caponigro, and T. Chambrion. “Which notion of energy for bilinear quantum systems?” In: *LHMNLC - 4th IFAC Workshop on Lagrangian and Hamiltonian Methods for Non Linear Control - 2012*. Bertinoro, Italy, Aug. 2012, pp. 226–230.
- [13] M. J. Bravo, M. Caponigro, E. Leibowitz, and B. Piccoli. “Keep right or left? Towards a cognitive-mathematical model for pedestrians”. In: *Networks and Heterogeneous Media* 10.3 (2015), pp. 559–578.
- [14] M. Caponigro, M. Fornasier, B. Piccoli, and E. Trélat. “Sparse stabilization and control of alignment models”. In: *Mathematical Models and Methods in Applied Sciences* 25.03 (2015), pp. 521–564.
- [15] M. Caponigro, M. Fornasier, B. Piccoli, and E. Trélat. “Sparse stabilization and optimal control of the Cucker-Smale model”. In: *Mathematical Control and Related Fields* 3.4 (2013), pp. 447–466.
- [16] M. Caponigro, R. Ghezzi, B. Piccoli, and E. Trélat. “Regularization of chattering phenomena via bounded variation controls”. Preprint hal-01359037. Sept. 2016.
- [17] M. Caponigro, A. C. Lai, and B. Piccoli. “A nonlinear model of opinion formation on the sphere”. In: *Discrete and Continuous Dynamical Systems* 35.9 (2015), pp. 4241–4268.
- [18] M. Caponigro, B. Piccoli, F. Rossi, and E. Trélat. “Mean-field sparse Jurdjevic–Quinn control”. In: *Mathematical Models and Methods in Applied Sciences* 27.07 (2017), pp. 1223–1253.
- [19] M. Caponigro, B. Piccoli, F. Rossi, and E. Trélat. “Sparse Feedback Stabilization of Multi-Agent Dynamics”. In: *Decision and Control (CDC), 2016 IEEE 55th Annual Conference on*. 2016.
- [20] M. Caponigro, B. Piccoli, F. Rossi, and E. Trélat. “Sparse Jurdjevic–Quinn stabilization of dissipative systems”. In: *Automatica* 86 (2017), pp. 110–120.
- [21] M. Caponigro and M. Sigalotti. “Exact controllability in projections of the bilinear Schrödinger equation”. Preprint hal-01509971. 2017.
- [22] S. Wongkaew, M. Caponigro, K. Kułakowski, and A. Borzì. “On the control of the Heider balance model”. In: *The European Physical Journal Special Topics* 224.17-18 (2015), pp. 3325–3342.
- [23] S. Wongkaew, M. Caponigro, and A. Borzì. “On the control through leadership of the Hegselmann-Krause opinion formation model”. In: *Mathematical Models and Methods in Applied Sciences* 25.03 (2015), pp. 565–585.

Other References

- [24] A. A. Agrachev and A. V. Sarychev. “Navier–Stokes equations: controllability by means of low modes forcing”. In: *Journal of Mathematical Fluid Mechanics* 7.1 (2005), pp. 108–152.

-
- [25] S. Ahn, H.-O. Bae, S.-Y. Ha, Y. Kim, and H. Lim. “Application of flocking mechanism to the modeling of stochastic volatility”. In: *Mathematical Models and Methods in Applied Sciences* 23.09 (2013), pp. 1603–1628.
 - [27] G. Albi, M. Bongini, E. Cristiani, and D. Kalise. “Invisible control of self-organizing agents leaving unknown environments”. In: *SIAM Journal on Applied Mathematics* 76.4 (2016), pp. 1683–1710.
 - [28] G. Albi and L. Pareschi. “Modeling of self-organized systems interacting with a few individuals: from microscopic to macroscopic dynamics”. In: *Applied Mathematics Letters* 26.4 (2013), pp. 397–401.
 - [29] L. Ambrosio and W. Gangbo. “Hamiltonian ODEs in the Wasserstein space of probability measures”. In: *Communications on Pure and Applied Mathematics* 61.1 (2008), pp. 18–53.
 - [30] G. Arechavaleta, J.-P. Laumond, H. Hicheur, and A. Berthoz. “An optimality principle governing human walking”. In: *IEEE Transactions on Robotics* 24.1 (2008), pp. 5–14.
 - [31] J. M. Ball, J. E. Marsden, and M. Slemrod. “Controllability for distributed bilinear systems”. In: *SIAM J. Control Optim.* 20.4 (1982), pp. 575–597.
 - [32] M. Ballerini, N. Cabibbo, R. Candelier, A. Cavagna, E. Cisbani, I. Giardina, V. Lecomte, A. Orlandi, G. Parisi, A. Procaccini, M. Viale, and V. Zdravkovic. “Interaction ruling animal collective behavior depends on topological rather than metric distance: Evidence from a field study”. In: *Proceedings of the National Academy of Sciences* 105.4 (2008), pp. 1232–1237.
 - [33] L. Baudouin, O. Kavian, and J.-P. Puel. “Regularity for a Schrödinger equation with singular potentials and application to bilinear optimal control”. In: *J. Differential Equations* 216.1 (2005), pp. 188–222.
 - [34] L. Baudouin and J. Salomon. “Constructive solution of a bilinear optimal control problem for a Schrödinger equation”. In: *Systems Control Lett.* 57.6 (2008), pp. 453–464.
 - [35] K. Beauchard. “Local controllability of a 1-D Schrödinger equation”. In: *J. Math. Pures Appl. (9)* 84.7 (2005), pp. 851–956.
 - [36] K. Beauchard and J.-M. Coron. “Controllability of a quantum particle in a moving potential well”. In: *J. Funct. Anal.* 232.2 (2006), pp. 328–389.
 - [37] K. Beauchard, J.-M. Coron, and H. Teismann. “Minimal time for the bilinear control of Schrödinger equations”. In: *Systems Control Lett.* 71 (2014), pp. 1–6.
 - [38] K. Beauchard and C. Laurent. “Local controllability of 1D linear and nonlinear Schrödinger equations with bilinear control”. In: *J. Math. Pures Appl. (9)* 94.5 (2010), pp. 520–554.
 - [39] K. Beauchard and M. Morancey. “Local controllability of 1D Schrödinger equations with bilinear control and minimal time”. In: *Math. Control Relat. Fields* 4.2 (2014), pp. 125–160.

-
- [40] N. Bellomo, D. Clarke, L. Gibelli, P. Townsend, and B. Vreugdenhil. “Human behaviours in evacuation crowd dynamics: From modelling to “big data” toward crisis management”. In: *Physics of Life Reviews* 18 (2016), pp. 1–21.
 - [41] N. Bellomo and A. Bellouquid. “On the modeling of crowd dynamics: Looking at the beautiful shapes of swarms”. In: *Networks and Heterogeneous Media* 6 (2011), pp. 383–399.
 - [42] N. Bellomo and C. Dogbe. “On the modeling of traffic and crowds: A survey of models, speculations, and perspectives”. In: *SIAM review* 53.3 (2011), pp. 409–463.
 - [43] N. Bellomo and L. Gibelli. “Toward a mathematical theory of behavioral-social dynamics for pedestrian crowds”. In: *Mathematical Models and Methods in Applied Sciences* 25.13 (2015), pp. 2417–2437.
 - [44] N. Bellomo and J. Soler. “On the mathematical theory of the dynamics of swarms viewed as complex systems”. In: *Mathematical Models and Methods in Applied Sciences* 22.sup01 (2012), p. 1140006.
 - [45] A. L. Bertozzi, J. A. Carrillo, and T. Laurent. “Blow-up in multidimensional aggregation equations with mildly singular interaction kernels”. In: *Nonlinearity* 22.3 (2009), p. 683.
 - [46] W. Bialek, A. Cavagna, I. Giardina, T. Mora, O. Pohl, E. Silvestri, M. Viale, and A. M. Walczak. “Social interactions dominate speed control in poising natural flocks near criticality”. In: *Proceedings of the National Academy of Sciences* 111.20 (2014), pp. 7212–7217.
 - [47] R. M. Bianchini and G. Stefani. “Time-optimal problem and time-optimal map”. In: *Rend. Sem. Mat. Univ. Politec. Torino* 48.3 (1990), 401–429 (1992).
 - [48] R. S. Bliss and D. Burgarth. “Quantum control of infinite-dimensional many-body systems”. In: *Phys. Rev. A* 89 (2014), p. 032309.
 - [49] A. M. Bloch, R. W. Brockett, and C. Rangan. “Finite controllability of infinite-dimensional quantum systems”. In: *IEEE Trans. Automat. Control* 55.8 (2010), pp. 1797–1805.
 - [50] V. D. Blondel, J. M. Hendrickx, and J. N. Tsitsiklis. “Continuous-time average-preserving opinion dynamics with opinion-dependent communications”. In: *SIAM Journal on Control and Optimization* 48.8 (2010), pp. 5214–5240.
 - [51] M. Bongini, M. Fornasier, F. Fröhlich, L. Haghverdi, et al. “Sparse stabilization of dynamical systems driven by attraction and avoidance forces”. In: *Networks and Heterogeneous Media* 9.1 (2014), pp. 1–31.
 - [52] A. Borzi and E. Decker. “Analysis of a leap-frog pseudospectral scheme for the Schrödinger equation”. In: *J. Comput. Appl. Math.* 193.1 (2006), pp. 65–88.
 - [53] A. Borzi and S. Wongkaew. “Modeling and control through leadership of a refined flocking system”. In: *Mathematical Models and Methods in Applied Sciences* 25.02 (2015), pp. 255–282.
 - [54] J. Bourgain. “On growth of Sobolev norms in linear Schrödinger equations with smooth time dependent potential”. In: *J. Anal. Math.* 77 (1999), pp. 315–348.

-
- [55] R. W. Brockett. “System theory on group manifolds and coset spaces”. In: *SIAM J. Control* 10 (1972), pp. 265–284.
 - [56] R. W. Brockett. “Asymptotic stability and feedback stabilization”. In: *Differential geometric control theory* 27.1 (1983), pp. 181–191.
 - [57] F. Bullo, J. Cortés, and S. Martinez. “Distributed control of robotic networks: a mathematical approach to motion coordination algorithms”. In: *Princeton series in applied mathematics*. Princeton University Press, Princeton (2009).
 - [58] J. A. Cañizo, J. A. Carrillo, and J. Rosado. “A well-posedness theory in measures for some kinetic models of collective motion”. In: *Mathematical Models and Methods in Applied Sciences* 21.03 (2011), pp. 515–539.
 - [59] C. Canuto, F. Fagnani, and P. Tilli. “An Eulerian Approach to the Analysis of Krause’s Consensus Models”. In: *SIAM Journal on Control and Optimization* 50.1 (2012), pp. 243–265.
 - [60] J. A. Carrillo, M. Fornasier, J. Rosado, and G. Toscani. “Asymptotic flocking dynamics for the kinetic Cucker-Smale model”. In: *SIAM J. Math. Anal.* 42.1 (2010), pp. 218–236.
 - [61] T. Chambrion, P. Mason, M. Sigalotti, and U. Boschain. “Controllability of the discrete-spectrum Schrödinger equation driven by an external field”. In: *Ann. Inst. H. Poincaré Anal. Non Linéaire* 26.1 (2009), pp. 329–349.
 - [62] F. Clarke. “Lyapunov functions and discontinuous stabilizing feedback”. In: *Annual reviews in control* 35.1 (2011), pp. 13–33.
 - [63] C. Cohen-Tannoudji, B. Diu, and F. Laloë. *Quantum mechanics*. Quantum Mechanics. Wiley, 1977.
 - [64] J.-M. Coron. *Control and nonlinearity*. 136. American Mathematical Soc., 2007.
 - [65] J.-M. Coron. “Global asymptotic stabilization for controllable systems without drift”. In: *Mathematics of Control, Signals and Systems* 5.3 (1992), pp. 295–312.
 - [66] I. Couzin, J. Krause, R. James, G. Ruxton, and N. Franks. “Collective memory and spatial sorting in animal groups.” In: *J Theor Biol* 218.1–11 (2002).
 - [67] E. Cristiani, P. Frasca, and B. Piccoli. “Effects of anisotropic interactions on the structure of animal groups”. In: *Journal of mathematical biology* 62.4 (2011), pp. 569–588.
 - [68] E. Cristiani, B. Piccoli, and A. Tosin. *Multiscale modeling of pedestrian dynamics*. Vol. 12. Springer, 2014.
 - [69] F. Cucker and J.-G. Dong. “Avoiding collisions in flocks”. In: *IEEE transactions on automatic control* 55.5 (2010), pp. 1238–1243.
 - [70] F. Cucker and S. Smale. “Emergent behavior in flocks”. In: *IEEE Trans. Automat. Control* 52.5 (2007), pp. 852–862.
 - [71] D. D’Alessandro. *Introduction to quantum control and dynamics*. English. Applied Mathematics and Nonlinear Science Series. Boca Raton, FL: Chapman, Hall/CRC., 2008.

- [72] G. Deffuant, D. Neau, F. Amblard, and G. Weisbuch. “Mixing beliefs among interacting agents”. In: *Advances in Complex Systems* 3.01n04 (2000), pp. 87–98.
- [73] J. C. Dittmer. “Diskrete nichtlineare Modelle der Konsensbildung”. In: *Diploma thesis Universität Bremen* (2000).
- [74] R. El Assoudi, J. P. Gauthier, and I. A. K. Kupka. “On subsemigroups of semisimple Lie groups”. In: *Ann. Inst. H. Poincaré Anal. Non Linéaire* 13.1 (1996), pp. 117–133.
- [75] S. Ervedoza and J.-P. Puel. “Approximate controllability for a system of Schrödinger equations modeling a single trapped ion”. In: *Ann. Inst. H. Poincaré Anal. Non Linéaire* 26.6 (2009), pp. 2111–2136.
- [76] R. Escobedo, A. Ibañez, and E. Zuazua. “Optimal strategies for driving a mobile agent in a “guidance by repulsion” model”. In: *Communications in Nonlinear Science and Numerical Simulation* 39 (2016), pp. 58–72.
- [77] M. Fornasier, B. Piccoli, and F. Rossi. “Mean-field sparse optimal control”. In: *Phil. Trans. R. Soc. A* 372.2028 (2014), p. 20130400.
- [78] J. R. P. French. “A formal theory of social power”. In: *Psychological Review* 63 (1956), pp. 181–194.
- [79] A. T. Fuller. “Relay control systems optimized for various performance criteria”. In: *Proceedings of the 1st World Congress IFAC, Moscow, 1960, Butterworths, London (1961)*, pp. 510–519.
- [80] J.-P. Gauthier and G. Bornard. “Controlabilité des systèmes bilinéaires”. In: *SIAM J. Control Optim.* 20.3 (1982), pp. 377–384.
- [81] M. C. D. Gennaro and A. Jadbabaie. “Decentralized control of connectivity for multi-agent systems”. In: *IEEE Conference on Decision and Control, New Orleans, LA, USA (2007)*.
- [82] I. Giardina. “Collective behavior in animal groups: theoretical models and empirical studies”. In: *Human Frontier Science Program Journal* 205–219 (2008).
- [83] S.-Y. Ha, T. Ha, and J.-H. Kim. “Emergent behavior of a Cucker-Smale type particle model with nonlinear velocity couplings”. In: *IEEE Trans. Automat. Control* 55.7 (2010), pp. 1679–1683.
- [84] S.-Y. Ha and E. Tadmor. “From particle to kinetic and hydrodynamic descriptions of flocking”. In: *Kinet. Relat. Models* 1.3 (2008), pp. 415–435.
- [85] R. Hegselmann and U. Krause. “Opinion dynamics and bounded confidence models, analysis, and simulation”. In: *Journal of Artificial Societies and Social Simulation* 5.3 (2002).
- [86] F. Heider. “Social perception and phenomenal causality”. In: *Psychological review* 51.6 (1944), p. 358.
- [87] F. Heider. *The psychology of interpersonal relations*. Psychology Press, 2013.

-
- [88] D. Helbing and P. Molnar. “Social force model for pedestrian dynamics”. In: *Physical review E* 51.5 (1995), p. 4282.
 - [89] A. Huth and C. Wissel. “The simulation of the movement of fish schools”. In: *Journal of Theoretical Biology* 156 (1992), pp. 365–385.
 - [90] K. Ito and K. Kunisch. “Asymptotic properties of feedback solutions for a class of quantum control problems”. In: *SIAM J. Control Optim.* 48.4 (2009), pp. 2323–2343.
 - [91] P. E. Jabin and S. Motsch. “Clustering and asymptotic behavior in opinion formation”. In: *Journal of Differential Equations* 257.11 (Dec. 2014), pp. 4165–4187.
 - [92] A. Jadbabaie, J. Lin, and A. S. Morse. “Coordination of groups of mobile autonomous agents using nearest neighbor rules”. In: *Automatic Control, IEEE Transactions on* 48.6 (2003), pp. 988–1001.
 - [93] V. Jurdjevic and I. Kupka. “Control systems on semisimple Lie groups and their homogeneous spaces”. In: *Ann. Inst. Fourier (Grenoble)* 31.4 (1981), pp. vi, 151–179.
 - [94] V. Jurdjevic and J. P. Quinn. “Controllability and stability”. In: *Journal of differential equations* 28.3 (1978), pp. 381–389.
 - [95] V. Jurdjevic and H. J. Sussmann. “Control systems on Lie groups”. In: *J. Differential Equations* 12 (1972), pp. 313–329.
 - [96] T. Kato. “Integration of the equation of evolution in a Banach space”. In: *J. Math. Soc. Japan* 5 (1953), pp. 208–234.
 - [97] J. Krause and G. Ruxton. “Living in groups”. In: *Oxford series in ecology and evolution. Oxford University Press, New York* (2002).
 - [98] U. Krause. “A discrete nonlinear and non—autonomous model of consensus formation”. In: *Elaydi S, Ladas G, Popenda J and Rakowski J (Eds.), Communications in Difference Equations, Amsterdam: Gordon and Breach Publ.* (2000), pp. 227–236.
 - [99] U. Krause. “Soziale Dynamiken mit vielen Interakteuren, Eine Problemskizze”. In: *Krause U and Stöckler M (Eds.) Modellierung und Simulation von Dynamiken mit vielen interagierenden Akteuren, Universität Bremen* (1997), pp. 37–51.
 - [100] K. Kułakowski, P. Gawronski, and P. Gronek. “The Heider balance: A continuous approach”. In: *International Journal of Modern Physics C* 16.05 (2005), pp. 707–716.
 - [101] I. Kupka. “The ubiquity of Fuller’s phenomenon”. In: *Nonlinear controllability and optimal control*. Vol. 133. Monogr. Textbooks Pure Appl. Math. New York: Dekker, 1990, pp. 313–350.
 - [102] C. K. Law and J. H. Eberly. “Arbitrary control of a quantum electro-magnetic field”. In: *Phys. Rev. Lett.* 76.7 (1996), pp. 1055–1058.
 - [103] N. Leonard. “Multi-agent system dynamics: Bifurcation and behavior of animal groups”. In: *Plenary paper IFAC Symposium on Nonlinear Control Systems, Toulouse, France.* (2013).

-
- [104] P. Mason and M. Sigalotti. “Generic controllability properties for the bilinear Schrödinger equation”. In: *Comm. Partial Differential Equations* 35.4 (2010), pp. 685–706.
 - [105] M. Mirrahimi. “Lyapunov control of a quantum particle in a decaying potential”. In: *Ann. Inst. H. Poincaré Anal. Non Linéaire* 26.5 (2009), pp. 1743–1765.
 - [106] M. Mirrahimi and P. Rouchon. “Controllability of quantum harmonic oscillators”. In: *IEEE Transactions on Automatic Control* 49.5 (2004), pp. 745–747.
 - [107] M. Morancey. “Explicit approximate controllability of the Schrödinger equation with a polarizability term”. In: *Math. Control Signals Systems* 25.3 (2013), pp. 407–432.
 - [108] L. Moreau. “Stability of continuous-time distributed consensus algorithms”. In: *Decision and Control, 2004. CDC. 43rd IEEE Conference on*. Vol. 4. IEEE. 2004, pp. 3998–4003.
 - [109] L. Moreau. “Stability of multiagent systems with time-dependent communication links”. In: *Automatic Control, IEEE Transactions on* 50.2 (2005), pp. 169–182.
 - [110] S. Motsch and E. Tadmor. “Heterophilous Dynamics Enhances Consensus”. In: *SIAM Review* 56.4 (2014), pp. 577–621.
 - [111] V. Nersesyan. “Global approximate controllability for Schrödinger equation in higher Sobolev norms and applications”. In: *Ann. Inst. H. Poincaré Anal. Non Linéaire* 27.3 (2010), pp. 901–915.
 - [112] V. Nersesyan. “Growth of Sobolev Norms and Controllability of the Schrödinger Equation”. English. In: *Communications in Mathematical Physics* 290 (1 2009), pp. 371–387.
 - [113] V. Nersesyan and H. Nersisyan. “Global exact controllability in infinite time of Schrödinger equation”. In: *J. Math. Pures Appl. (9)* 97.4 (2012), pp. 295–317.
 - [114] G. Nicolis and I. Prigogine. “Self-Organization in Nonequilibrium Systems: From Dissipative Structures to Order Through Fluctuations”. In: *NY: Wiley* (1977).
 - [115] R. Olfati-Saber, J. A. Fax, and R. M. Murray. “Consensus and cooperation in networked multi-agent systems”. In: *Proceedings of the IEEE* 95.1 (2007), pp. 215–233.
 - [116] J. Parrish, S. Viscido, and D. Grunbaum. “Self-organized fish schools: an examination of emergent properties”. In: *The Biological Bulletin* 202 (2002), pp. 296–305.
 - [117] L. Perea, P. Elosegui, and G. Gómez. “Extension of the Cucker–Smale Control Law to Space Flight Formations”. In: *Journal of Guidance, Control, and Dynamics* 32 (6 2009), pp. 527–537.
 - [118] B. Piccoli and F. Rossi. “Transport equation with nonlocal velocity in Wasserstein spaces: convergence of numerical schemes”. In: *Acta applicandae mathematicae* 124.1 (2013), pp. 73–105.
 - [119] B. Piccoli, F. Rossi, and E. Trélat. “Control to Flocking of the Kinetic Cucker–Smale Model”. In: *SIAM Journal on Mathematical Analysis* 47.6 (2015), pp. 4685–4719.
 - [120] I. Prigogine and I. Stengers. “Order out of Chaos. 1984”. In: *Bantam: New York* 108 (1984).

-
- [121] Y. Privat and M. Sigalotti. “The squares of the Laplacian-Dirichlet eigenfunctions are generically linearly independent”. In: *ESAIM Control Optim. Calc. Var.* 16.3 (2010), pp. 794–805.
 - [122] M. Reed and B. Simon. *Methods of modern mathematical physics. II. Fourier analysis, self-adjointness*. New York: Academic Press [Harcourt Brace Jovanovich Publishers], 1975, pp. xv+361.
 - [123] M. Reed and B. Simon. *Methods of modern mathematical physics. IV. Analysis of operators*. Academic Press [Harcourt Brace Jovanovich, Publishers], New York-London, 1978, pp. xv+396.
 - [124] H. Robbins. “Junction phenomena for optimal control with state-variable inequality constraints of third order”. In: *J. Optim. Theory Appl.* 31.1 (1980), pp. 85–99.
 - [125] L. Rosier and J.-M. Coron. “A relation between continuous time-varying and discontinuous feedback stabilization”. In: *J. Math. Syst. Estim. Control* 4 (1994), pp. 67–84.
 - [126] P. Rouchon. “Quantum systems and control”. In: *ARIMA Rev. Afr. Rech. Inform. Math. Appl.* 9 (2008), pp. 325–357.
 - [127] Y. L. Sachkov. “Controllability of invariant systems on Lie groups and homogeneous spaces”. In: *J. Math. Sci. (New York)* 100.4 (2000). Dynamical systems, 8, pp. 2355–2427.
 - [128] C. Samson. “Velocity and torque feedback control of a nonholonomic cart”. In: *Advanced Robot Control: Proceedings of the International Workshop on Nonlinear and Adaptive Control: Issues in Robotics, Grenoble, France, Nov. 21–23, 1990*. Ed. by C. Canudas de Wit. Berlin, Heidelberg: Springer Berlin Heidelberg, 1991, pp. 125–151.
 - [129] C. Sayrin, I. Dotsenko, X. Zhou, B. Peaudecerf, T. Rybarczyk, S. Gleyzes, P. Rouchon, M. Mirrahimi, H. Amini, M. Brune, J.-M. Raimond, and S. Haroche. “Real-time quantum feedback prepares and stabilizes photon number states”. In: *Nature* 477 (2011), pp. 73–77.
 - [130] S. G. Schirmer, H. Fu, and A. I. Solomon. “Complete controllability of quantum systems”. In: *Physical Review A* 63.6 (2001), p. 063410.
 - [131] Y. Sharon and M. Margaliot. “Third-order nilpotent, nice reachability and asymptotic stability”. In: *J. Differ. Equations* 233 (2007), pp. 136–150.
 - [132] J. Shen. “Cucker-Smale flocking under hierarchical leadership”. In: *SIAM Journal on Applied Mathematics* 68.3 (2007), pp. 694–719.
 - [133] A. Shirikyan. “Exact controllability in projections for three-dimensional Navier–Stokes equations”. In: *Annales de l’Institut Henri Poincaré (C) Non Linear Analysis* 24.4 (2007), pp. 521–537.
 - [134] P. Sobkowicz. “Modelling Opinion Formation with Physics Tools: Call for Closer Link with Reality”. In: *Journal of Artificial Societies and Social Simulation* 12.1 (2009), p. 11.

- [135] E. Sontag and H. Sussmann. “Remarks on continuous feedback”. In: *Decision and Control including the Symposium on Adaptive Processes, 1980 19th IEEE Conference on*. Dec. 1980, pp. 916–921.
- [136] E. D. Sontag. *Mathematical control theory: deterministic finite dimensional systems*. Vol. 6. Springer Science & Business Media, 2013.
- [137] D. Sumpter. “The principles of collective animal behaviour.” In: *Philosophical Transaction of the Royal Society B* 361 (2006), pp. 5–22.
- [138] H. J. Sussmann. “A bang-bang theorem with bounds on the number of switchings”. In: *SIAM J. Control Optim.* 17.5 (1979), pp. 629–651.
- [139] K. Sznajd-Weron and J. Sznajd. “Opinion evolution in closed community”. In: *International Journal of Modern Physics C* 11.06 (2000), pp. 1157–1165.
- [140] H. G. Tanner, A. Jadbabaie, and G. J. Pappas. “Flocking in fixed and switching networks”. In: *IEEE Trans. Automat. Control* 52.5 (2007), pp. 863–868.
- [141] G. Turinici. “On the controllability of bilinear quantum systems”. In: *Mathematical models and methods for ab initio Quantum Chemistry*. Ed. by M. Defranceschi and C. Le Bris. Vol. 74. Lecture Notes in Chemistry. Springer, 2000.
- [142] M. I. Zelikin and V. F. Borisov. *Theory of chattering control*. Systems & Control: Foundations & Applications. Boston, MA: Birkhäuser Boston Inc., 1994, pp. xvi+242.
- [143] J. Zhang, K. H. Johansson, J. Lygeros, and S. Sastry. “Zeno hybrid systems”. In: *International Journal of Robust and Nonlinear Control* 11.5 (2001), pp. 435–451.
- [144] E. Zuazua. “Remarks on the controllability of the Schrödinger equation”. In: *Quantum control: mathematical and numerical challenges* 33 (2003), pp. 193–211.